

FEATURES

- 256-channel, charge-to-digital converter**
- 16-bit resolution with no missing codes**
- Simultaneous sampling**
- User adjustable, forward charge range up to 16.0 pC**
- Down to 60 μ s line time**
- Ultralow noise: 580 e^- rms at 2 pC range**
- INL: ± 2.5 LSBs or 38.125 ppm, ADC included**
- Very low power dissipation**
- Multiple functional power modes**
 - Low power mode: 1 mW per channel**
 - Normal mode: 1.4 mW per channel**
- Multiple power-down and sleep modes, down to 0.005 mW per channel**
- Capable of measuring electron or hole collected charge**
- Tested and delivered with high density flex connections, SOF**
- LVDS/CMOS self clocked serial interface**
- SPI daisy-chain configuration registers**
- On-board AFE timing sequencer**
- On-board temperature sensor and reference buffer**

APPLICATIONS

- Digital x-ray panel readouts**
- Photodiode sensor arrays**
- CT scanners**
- High channel count data acquisition systems**
(current or voltage inputs)

GENERAL DESCRIPTION

The AD71143 is a 256-channel, charge-to-digital analog front end (AFE) mounted on high density flex. It can be directly mounted on a digital x-ray panel and is suitable for a large variety of digital x-ray or photodiode sensor array applications, including direct conversion hole and electron sensing and dynamic imaging. The AD71143 contains 256 low power, low noise, low input current integrators with a correlated double sampling stage; simultaneous sample-and-hold; and high speed, high resolution AFEs with various sampling rates and 16-bit resolution.

All converted channel results are output on a single low voltage differential signaling (LVDS), self clocked serial interface that significantly reduces external hardware.

A serial peripheral interface (SPI) compatible serial interface allows configuration of the AFE using the SDI input. The SDO output allows the user to daisy-chain several AFEs on a single 3-wire bus.

An integrated AFE timing sequencer controls the sampling activity of the AFE. The sequencer is programmed via the SPI port and is timed by a single clock, ACLK.

FUNCTIONAL BLOCK DIAGRAM

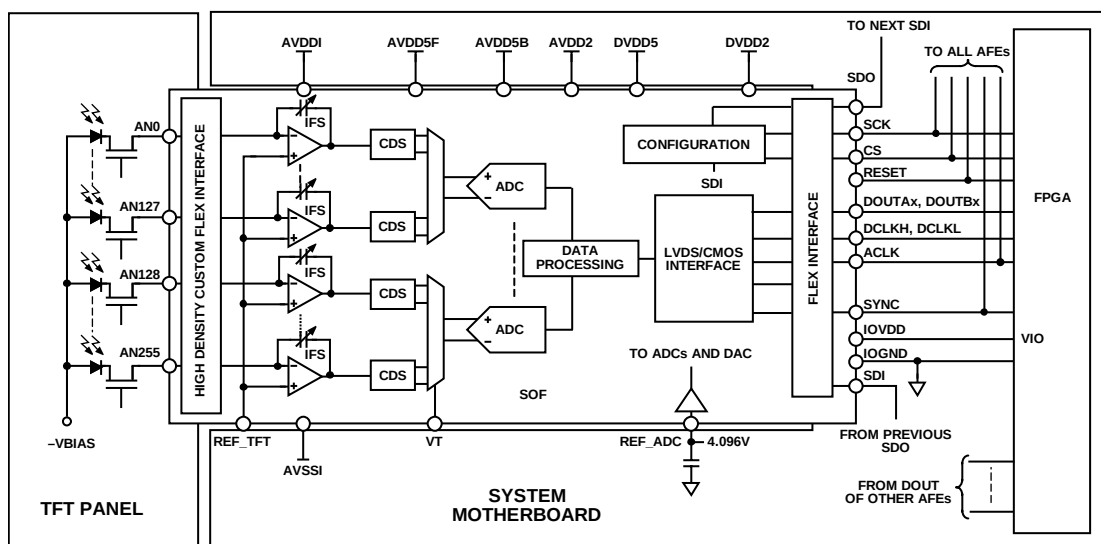


Figure 1.

Rev. SpA

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REVISION HISTORY

5/2016—Rev. Sp0 to Rev. SpA

Change to Features Section	1	Changes to Figure 22.....	17
Changes to CDS1 Open, CDS2 Open to DCLKH/DCLKL Delay Parameter and RESET to CS Rising Edge Parameter, Table 2	7	Added Figure 23; Renumbered Sequentially	18
Changes to Table 3.....	8	Changes to Figure 29.....	21
Change to Figure 15	13	Changes to Figure 38.....	26
Change to Figure 19	15	Changes to Table 12	31
Changes to AFE Sequencer Timing Section	16		

10/2014—Revision Sp0: Initial Version

SPECIFICATIONS

AVDD5F = AVDD5B = AVDD5Q = DVDD5 = 5 V, AVDD2 = DVDD2 = 2.5 V, AVDDI = REF_TFT + 1.2 V, AVSSI = REF_TFT - 0.6 V, IOVDD = 2.5 V, REF_TFT = 1 V to 2.7 V, REF_ADC = 4.096 V, $\tau_{LPF} = 3.9 \mu s$, $t_{LINE} = 60 \mu s$, $C_F = 0.5 pF$, $C_{SOURCE} = 50 pF$, $T_J = 50^\circ C$, unless otherwise noted.

Table 1.

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
ANALOG INPUT					
Forward Charge Range (FSR)	Nominal range, $0 \leq IFS^1 \leq 31$	0.5	$0.5 \times (IFS + 1)$	16.0	pC
Integrator Feedback Capacitance (C_F)	Nominal range, $0 \leq IFS^1 \leq 31$	0.125	$0.125 \times (IFS + 1)$	4	pF
Charge Range Accuracy			± 5		%
Charge Range Channel-to-Channel Matching	Within a chip		± 0.5		%
	From chip to chip		± 5		%
Reverse Charge Range			See Figure 31		% FSR
Integrator Common-Mode Voltage (REF_TFT)		0.6		2.7	V
Source Capacitance ²		0		200	pF
Integrator Output Transconductance			4		mS
Input/Output Current	Integrator source or sink current				
	Normal mode		± 40		μA
	Low power mode		± 10		μA
	Sleep mode		± 2.5		μA
THROUGHPUT					
Line Time (t_{LINE}) ³	$\tau_{LPF} = 1.3 \mu s$	60			μs
Low-Pass Filter Time Constant (τ_{LPF})	Nominal range, $0 \leq LPF^4 \leq 3$, ADCAVG ⁵ = 0	1.3		11.7	μs
Low-Pass Filter Time Constant Accuracy			± 8		%
Low-Pass Filter Time Constant Drift			1250		ppm/ $^\circ C$
ACCURACY					
Resolution			16		Bits
Integral Nonlinearity (INL)	REF_TFT + 8000 LSB		± 2.5		LSB
Integral Nonlinearity Match	Channel-to-channel across chips over full ADC range		± 6		LSB
Crosstalk	Adjacent channels		0.07		%
	All other channels		0.01		%
Input Referred Random Noise	Normal mode, $\tau_{LPF} = 3.9 \mu s$, $t_{LINE} = 60 \mu s$, $C_F = 0.5 pF$, ADCAVG ⁵ = 0		750		e ⁻ rms
	Low power mode, $\tau_{LPF} = 3.9 \mu s$, $t_{LINE} = 60 \mu s$, $C_F = 0.5 pF$, ADCAVG ⁵ = 0		880		e ⁻ rms
	Normal mode, $\tau_{LPF} = 3.9 \mu s$, $t_{LINE} = 100 \mu s$, $C_F = 0.5 pF$, ADCAVG ⁵ = 1		580		e ⁻ rms
	Low power mode, $\tau_{LPF} = 3.9 \mu s$, $t_{LINE} = 100 \mu s$, $C_F = 0.5 pF$, ADCAVG ⁵ = 1		730		e ⁻ rms
Input Referred Correlated Noise Ratio to Random Noise ⁶	$C_{SOURCE} = 0 pF$		1:6		
Input Offset Voltage	Measured with respect to REF_TFT		± 2		mV
Input Offset Leakage Current			± 1		pA
Input Offset Current Drift	$T_J = 25^\circ C$ to $45^\circ C$		20		fA/ $^\circ C$
Input Offset Current Stability			0.25		fA/min
Range Sensitivity to REF_ADC			1:1		

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
Gain Drift			25		ppm/°C
Power Supply Sensitivity	AVDDI – AVSSI = 2.5 V at 10 kHz		40		dB
	AVDDI – AVSSI = 1.8 V at 10 kHz		20		dB
ADC REFERENCE (REF_ADC)					
Input Reference Voltage Range	See the ADR444 data sheet	4.091	4.096	4.101	V
Input Reference Current			10		nA
VOLTAGE TEST INPUT (VT) ⁷					
Analog Voltage Range		0.5		4.5	V
Input Impedance			2.5		MΩ
Resolution			16		Bits
Integral Linearity Error			±10		LSB
Transition Noise			2		LSB
TEMPERATURE SENSOR (TEMP)					
Resolution			16		Bits
Temperature Sensitivity			17		LSB/°C
DIGITAL INPUTS					
CMOS Logic Levels	CMOS inputs are all 5 V tolerant				
Input Voltage					
Low (V _{IL})		–0.3		+0.6	V
High (V _{IH})		2.0		5.25	V
Input Current					
Low (I _{IL})		–1		+1	μA
High (I _{IH})		–1		+1	μA
DIGITAL OUTPUTS					
Output Voltage					
Low (V _{OL})	I _{SINK} = 500 μA, SDO			0.4	V
High (V _{OH})	I _{SOURCE} = –500 μA, SDO	IOVDD – 0.3			V
LVDS INPUT/OUTPUT					
Data Format			Serial LVDS straight binary		
Differential Output Voltage (V _{OD}) ⁸	R _L = 100 Ω	225	350	400	mV
Common-Mode Output Voltage (V _{OCM}) ⁸	R _L = 100 Ω	1000	1250	1300	mV
Differential Input Voltage (V _{ID})		100		650	mV
Common-Mode Input Voltage (V _{ICM})		100		1575	mV
Differential Input Impedance	DCLKH/DCLKL inputs		100		Ω
POWER MODE SWITCHING TIMES					
Enable Time	Into sleep		0.1		ms
	Into power-down (REFDACDIS ⁹ is low) ¹⁰		1.0		ms
Disable Time	Out of sleep		0.1		ms
	Out of power-down (REFDACDIS ⁹ is low) ¹⁰		0.1		ms
POWER SUPPLIES ¹¹					
AVDD2, DVDD2		2.375	2.5	2.625	V
AVDD5F, AVDD5B, DVDD5		4.75	5	5.25	V
AVDD5Q		AVDD5F – 0.2	AVDD5F	5.25	V
AVDDI		REF_TFT + 1.2			V
AVSSI				REF_TFT – 0.6	V
AVDDI to AVSSI		1.8		2.7	V
AVDD5F to AVSSI		1.8		5.25	V
AVDD5Q to AVDDI		–0.2			V
IOVDD		2.3	2.5	2.7	V

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
SUPPLY CURRENT ¹²					
Power-Down Mode					
IOVDD			50		μA
DVDD2			35		μA
DVDD5			75		μA
AVDD2			15		μA
AVDD5B			30		μA
AVDD5F			65		μA
AVDD5Q			10		μA
AVDDI			85		μA
AVSSI			−10		μA
Sleep Mode					
IOVDD			50		μA
DVDD2			35		μA
DVDD5			110		μA
AVDD2			300		μA
AVDD5B			850		μA
AVDD5F			3.3		mA
AVDD5Q			10		μA
AVDDI			22		mA
AVSSI			−22		mA
Low Power Mode					
IOVDD			10		mA
DVDD2			710		μA
DVDD5			70		μA
AVDD2			8.5		mA
AVDD5B			28		mA
AVDD5F			3.5		mA
AVDD5Q			10		μA
AVDDI			21		mA
AVSSI			−21		mA
Normal Mode					
IOVDD			17	22.5	mA
DVDD2			600		μA
DVDD5			200		μA
AVDD2			9.5	12	mA
AVDD5B			27	37	mA
AVDD5F			8.5	10	mA
AVDD5Q			10		μA
AVDDI			65	80	mA
AVSSI		−80	−65		mA
Power Dissipation					
Power-Down Mode					
	REFDACDIS ¹³ is high, PDTFTEN ¹⁴ is high		1.2		mW
			16		mW
	REFDACDIS is low, PDTFTEN ¹⁴ is high		21		mW
Sleep Mode					
	REFDACDIS ¹³ is high		56		mW
	REFDACDIS is low		61		mW
Low Power Mode					
			260		mW
Normal Mode					
			363		mW

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
Power Dissipation per Channel					
Power-Down Mode	REFDACDIS is low, PDTFTEN ¹⁴ is high		0.005		mW/ch
	REFDACDIS ¹³ is high, PDTFTEN ¹⁴ is high		0.082		mW/ch
			0.063		mW/ch
Sleep Mode	REFDACDIS ¹³ is high		0.22		mW/ch
	REFDACDIS is low		0.24		mW/ch
Low Power Mode			1.02		mW/ch
Normal Mode			1.42		mW/ch
TEMPERATURE RANGE	T _J = 50°C	0		70	°C

¹ The five integrator full-scale (IFS) selection bits in Configuration Register 0 allow the user to select the full-scale range of the integrator by adjusting the internal feedback capacitor.

² Source capacitances > 200 pF are feasible with slower AFE timing.

³ The minimum line time in pipeline mode is 70 μs in single data rate mode and 60 μs in double data rate mode.

⁴ The two low-pass filter time constant bits (LPF) in Configuration Register 1 allow the user to select the time constant of the low-pass filter at the output of the integrator. For 0 μs, filter acceleration must be programmed via Configuration Register 7. See the AFE Operation section for more details.

⁵ ADCAVG (Bit 1 of Configuration Register 1) enables ADC averaging. If enabled, the low-pass filter time constants double. For example, 3 μs becomes 6 μs. See the bit description in Table 12 for more information.

⁶ Average noise is equal to the root-sum square (RSS) of the correlated noise and random noise. Correlated noise can be calculated by averaging the results of all channels in a given sample. The standard deviation of these average results across N number of samples gives the correlated noise. The average noise can be calculated as the standard deviation of each individual channel averaged together. The random noise is determined by de-RSSing the correlated noise from the average noise.

⁷ The output code of VT (VT_{CODE}) can be translated to a voltage using the following formula: VT = (VT_{CODE} × 61.035 μV) + 0.5 V. For example, if VT = 0.5 V, VT_{CODE} = 0 LSB.

⁸ The ANSI-644 specification for V_{OD} is 247 mV minimum and 454 mV maximum; for V_{OCM}, it is 1125 mV minimum and 1375 mV maximum.

⁹ Keeping the internal reference DAC enabled allows the fastest switching time because all capacitors that are connected to REF_OUT are biased to their final voltage. If the reference DAC is disabled by setting REFDACDIS (Bit 7 of Configuration Register 2) high, the switching time is dominated by the time constant that is associated with the capacitance seen by REF_OUT.

¹⁰ Power mode switching times into and out of these modes are longer when REFDACDIS is high because REF_OUT is used to charge and discharge any external capacitance connected to REF_OUT.

¹¹ See the Power Supply Restrictions and Sequencing section for constraints with the AFE power supplies.

¹² See the Power Optimization (Sleep Mode and Power-Down Modes) section for details.

¹³ When REFDACDIS (Bit 7 of Configuration Register 2) is high, the on-board reference DAC is disabled. See the Power Optimization (Sleep Mode and Power-Down Modes) section for details.

¹⁴ When PDTFTEN (Bit 8 of Configuration Register 2) is high, the input stage of the integrator is biased to REF_TFT in power-down mode. See the Power Optimization (Sleep Mode and Power-Down Modes) section for details.

TIMING SPECIFICATIONS

$T_j = 0^\circ\text{C}$ to 70°C , IOVDD = 2.3 V to 2.7 V, unless otherwise noted.

Table 2.

Parameter	Symbol	Min	Typ	Max	Unit
Burst Duration	t_{BURST}	1765	$t_{\text{DCLK}} \times (64 + M^1)$		ns
DCLKH/DCLKL LVDS Clock Period	t_{DCLK}	5		20	ns
DCLKH/DCLKL Frequency	f_{DCLK}	50		200	MHz
DCLKH/DCLKL to DOUTxH/DOUTxL Delay	t_{DDOUT}	0	4	7	ns
DCLKH/DCLKL to DCLKOH/DCLKOL Delay (Echo Clock Mode)	t_{DCO}	0	4	7	ns
DCLKOH/DCLKOL to DOUTxH/DOUTxL Delay (Echo Clock Mode)	t_{D}	-1	0	+1	ns
CDS1 Open, CDS2 Open to DCLKH/DCLKL Delay	t_{DAFEDCLK}	1640			ns
SYNC to DCLKH/DCLKL Rising or Falling Edge	t_{SYNCDCLK}	20			ns
SYNC to ACLK Rising or Falling Edge	t_{SYNCACLK}	20			ns
SYNC High Time	t_{SYNCH}	10			ns
ACLK Clock Period	t_{ACLK}	20			ns
SCK SPI Clock Period	t_{SCK}	20			ns
SCK High Time	t_{SCKH}	10			ns
SCK Low Time	t_{SCKL}	10			ns
SCK Rising Edge to SDO Valid Delay	t_{DSDO}	0		15	ns
SDI Valid Setup Time from SCK Falling Edge	t_{SSDISCK}	5			ns
SDI Valid Hold Time from SCK Falling Edge	t_{HSDISCK}	0			ns
CS Valid Setup Time from SCK Rising Edge	t_{SCSSCK}	10			ns
CS Valid Hold Time from SCK Falling Edge	t_{HCSSCK}	10			ns
RESET High Time	t_{RESET}	10			ns
RESET to CS Rising Edge	t_{RESETCS}	4000			ns

¹ M is the number of muted clocks in the burst pattern (see the Conversion Operation section for more details).

ABSOLUTE MAXIMUM RATINGS

Table 3.

Parameter	Rating
Analog Inputs	
All ANx ¹ Pins	±0.3 V to REF_TFT
REF_TFT to AGND ²	−0.3 V to +2.3 V
REF_TFT to AVSSI	−0.3 V to +2.3 V
REF_TFT to AVDDI	−0.3 V to +3 V
REF_INT to AGND ²	−0.3 V to AVDD5B + 0.3 V
REF_ADC to ADC GND ²	−0.3 V to AVDD5B + 0.3 V
VT to AGND ²	−0.3 V to AVDD5B + 0.3 V
Analog Input Current	
All ANx ¹ Pins	±1.5 mA
Supply Voltage	
AVDDI to AGND ²	−0.3 V to +5.3 V
AVSSI to AGND ²	−0.3 V to +2.3 V
AVDDI to AVSSI	−0.3 V to +3 V
AVDD5F, AVDD5Q to AGND ²	−0.3 V to +6 V
AVDD5B, DVDD5 to ADC GND ²	−0.3 V to +6 V
AVDD2, DVDD2 to ADC GND ²	−0.3 V to +3 V
IOVDD to IOGND (Digital GND) ²	−0.3 V to +3 V
Digital Inputs to IOGND (Digital GND) ²	−0.3 V to +6 V
LVDS Inputs to IOGND (Digital GND) ²	−0.3 V to IOVDD + 0.3 V
Digital Outputs to IOGND (Digital GND) ²	−0.3 V to IOVDD + 0.3 V
Storage Conditions	
Factory Sealed	20°C ± 15°C, 50% ± 20% relative humidity (RH), 1 year
Unsealed/Opened	25°C ± 3°C, 50% ± 10% RH, 3 months
Nitrogen Purged Dry Box	25°C ± 3°C, 30% maximum RH, 6 months

¹ ANx pins include AN0 through AN255.² See Figure 17 for the ground and shield connection diagram. Note that AGND = AGND5Q = AGND5F, ADC GND = AGND5B = AGND2 = DGND2 = DGND5, and digital GND = IOGND.

Stresses at or above those listed under Absolute Maximum Ratings may cause permanent damage to the product. This is a stress rating only; functional operation of the product at these or any other conditions above those indicated in the operational section of this specification is not implied. Operation beyond the maximum operating conditions for extended periods may affect product reliability.

ESD CAUTION



ESD (electrostatic discharge) sensitive device.

Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

PIN CONFIGURATION AND FUNCTION DESCRIPTIONS

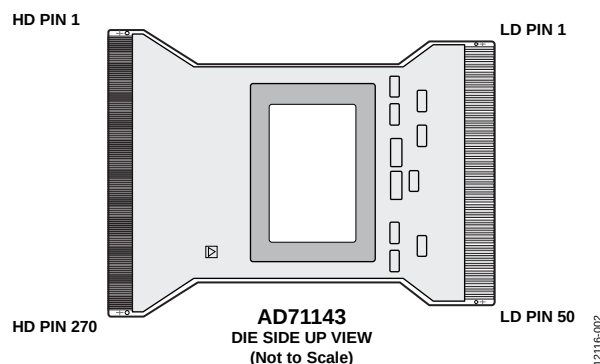


Figure 2. Pin Configuration

Table 4. Pin Function Descriptions

Pin No. ¹	Mnemonic	Type ²	Description
HD8 to HD263	AN0 to AN255	AI	Analog Charge Inputs. Connect these pins to the thin film transistor (TFT) panel.
LD10	VT	AI	Voltage Test Input. This pin provides a direct connection to the internal ADC.
LD12	REF_TFT	AI	Integrator Positive Input Voltage. The voltage range is from 0.6 V to 2.7 V and is defined by the panel requirements of the user. This is a sensitive node that requires good decoupling. It can be driven from REF_OUT on the system motherboard.
LD15	REF_ADC	AI	Reference Voltage Input. A high impedance input at 4.096 V nominal, using a single ADR444 .
LD13	REF_INT	AI	Optional External Integrator Reset Voltage. If the integrator is set to a voltage other than REF_TFT, apply a voltage to REF_INT and set EXTRST (Bit 2 of Configuration Register 1) high. The voltage range is from 0.5 V to 4.5 V. If REF_INT is unused, it must be tied to REF_OUT.
LD14	REF_OUT	AO	Reference Output Voltage. This voltage is the output of the on-board reference digital-to-analog converter (DAC), which has a range from 0.5 V to 4.5 V. The DAC is controlled via REFDAC (Bits[7:0] of Configuration Register 3). The REF_OUT pin supplies the REF_INT and REF_TFT voltages. This node requires good decoupling on the system motherboard. It is recommended that this decoupling be tied to AGND5Q before connecting to AGND (see Figure 32).
LD22	RESET	DI	Asynchronous Reset CMOS Input. When high, this input resets the digital circuitry and initializes the configuration to its default value. It must be applied after each power-up.
LD24	SYNC	DI	Conversion Synchronization CMOS Input. With DCLKH/DCLKL and ACLK, SYNC defines the start of a line conversion.
LD33, LD34	DOUTAH, DOUTAL	DO	Conversion Data Self Clocked LVDS Outputs A. The conversion data is output serially on these pins. The data-words are output per DCLKH/DCLKL cycle.
LD30, LD31	DOUTBH, DOUTBL	DO	Conversion Data Self Clocked LVDS Outputs B. These outputs are used only in double data rate mode, which is enabled by the DOUTMODE bit (Bit 2 of Configuration Register 2).
LD39, LD40	DCLKH, DCLKL	DI	Conversion Data LVDS Clock Inputs. This clock controls the internal conversion process and shifts out the conversion results. It is recommended that DCLKH/DCLKL be kept inactive when SCK is active. DCLKH/DCLKL have an internal 100 Ω termination resistor.
LD23	ACLK	DI	Analog Front End (AFE) Clock. This CMOS clock controls the AFE timing sequencer, which defines the sampling of the AFE.
LD36, LD37	DCLKOH, DCLKOL	DO	Optional Echo of the Conversion Data LVDS Clock. These clock outputs are a duplicate of DCLKH/DCLKL and can be used for data readout synchronization of multiple AD71143 devices. They are enabled by the ECHOCCLK bit (Bit 1 of Configuration Register 2).
LD25	CS	DI	Configuration Write CMOS Input. When CS is high, the contents of the configuration registers are shifted in and out by SCK. The configuration register is double-buffered and is updated only on the falling edge of CS. CS must be kept low during conversion.
LD26	SCK	DI	Configuration Serial Data Clock CMOS Input. SCK pulses are ignored when CS is low. It is recommended that SCK be kept low during conversion (when ACLK or DCLKH/DCLKL is active).
LD28	SDI	DI	Configuration Serial Data CMOS Input. SDI is used as a data input of the configuration registers. This pin allows the user to daisy-chain the contents of the configuration registers of many AD71143 devices onto a single serial line. The digital data level on SDI is sampled on the SCK falling edge.

Pin No. ¹	Mnemonic	Type ²	Description
LD27	SDO	DO	Configuration Serial Data CMOS Output. The 16 bits of the configuration register are output on this pin. This pin is synchronized to SCK and output on the SCK rising edge.
LD4	AVDDI	P	Integrator High Power Supply. This pin is the high supply voltage for the integrator, which has a minimum voltage of REF_TFT + 1.2 V. The difference between AVDDI and AVSSI must be in the range of 1.8 V to 2.7 V.
LD5	AVSSI	P	Integrator Low Power Supply. This pin is the low supply voltage for the integrator, which has a maximum voltage of REF_TFT – 0.6 V. The difference between AVDDI and AVSSI must be in the range of 1.8 V to 2.7 V. If REF_TFT is low enough, AVSSI can be connected to AGND (see Figure 17).
LD8	AVDD5F	P	Analog Front-End Power Supply, 5 V Nominal.
LD6	AVDD5Q	P	Quiet Analog Power Supply, 5 V Nominal. This connection usually comes from AVDD5B.
LD20	AVDD5B	P	Analog Back-End Power Supply, 5 V Nominal.
LD17	AVDD2	P	Analog Power Supply, 2.5 V Nominal.
LD47	DVDD5	P	Digital Power Supply, 5 V Nominal.
LD45	DVDD2	P	Digital Power Supply, 2.5 V Nominal.
LD43	IOVDD	P	Digital Input/Output Power Supply, 2.5 V Nominal.
LD9	AGND5F	P	High Voltage Front-End Analog Power Supply Ground.
LD19	AGND5B	P	High Voltage Back-End Analog Power Supply Ground.
LD7	AGND5Q	P	Quiet Analog Power Supply Ground.
LD18	AGND2	P	Low Voltage Analog Power Supply Ground.
LD46	DGND5	P	Digital Power Supply Ground.
LD44	DGND2	P	Digital Power Supply Ground.
LD42	IOGND	P	Digital I/O Power Supply Ground.
HD7, HD264	GUARD1, GUARD2	AO	Guard. These are traces driven to the same potential as REF_TFT. They do not need to be connected at the panel.
HD1 to HD4, LD1	PASS-THRU1	N/A	Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU1 connections is variable.
HD5, LD2	PASS-THRU2	N/A	Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU2 connections is variable.
HD6, LD3	PASS-THRU3	N/A	Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU3 connections is variable.
HD265, LD48	PASS-THRU4	N/A	Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU4 connections is variable.
HD266, LD49	PASS-THRU5		Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU5 connections is variable.
HD267 to HD270, LD50	PASS-THRU6	AI	Pass-Through Connections. These connections can be used to pass through a signal or voltage level from the TFT to the x-ray panel motherboard. These connections do not connect to the die. The number of PASS-THRU6 connections is variable.
LD11, LD16, LD21, LD29, LD32, LD35, LD38, LD41	SHIELD1 to SHIELD8	AI	Shield Connections. These connections serve multiple purposes. They isolate references, analog supplies, and digital inputs from each other. They also provide impedance matching for the digital LVDS inputs/outputs of the AD71143 on the flex substrate. These connections do not connect to the die. It is recommended that SHIELD1 be tied to AGND5Q, SHIELD2 be tied to the ground of the external reference, SHIELD3 be tied to AGND5B, and SHIELD4 to SHIELD8 be tied to IOGND.

¹ Contact Analog Devices, Inc., for the most up-to-date package outline drawing.

² AI is analog input, AO is analog output, DI is digital input, DO is digital output, P is power, and N/A is not applicable.

TYPICAL PERFORMANCE CHARACTERISTICS

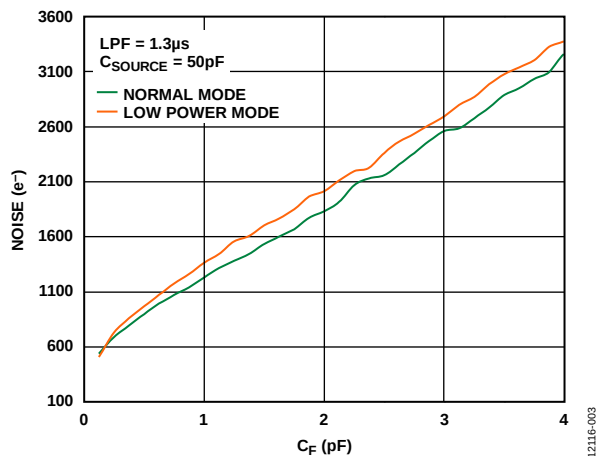
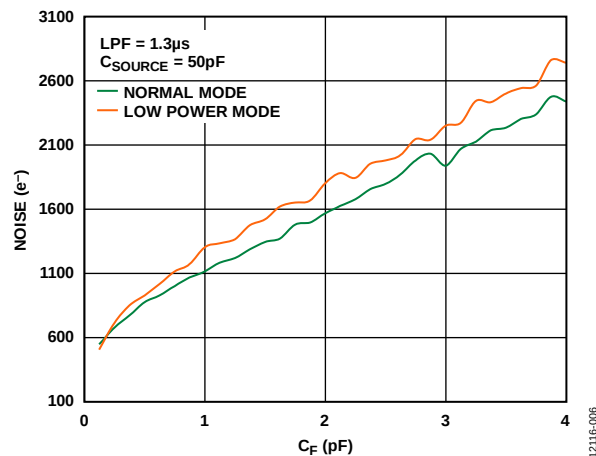
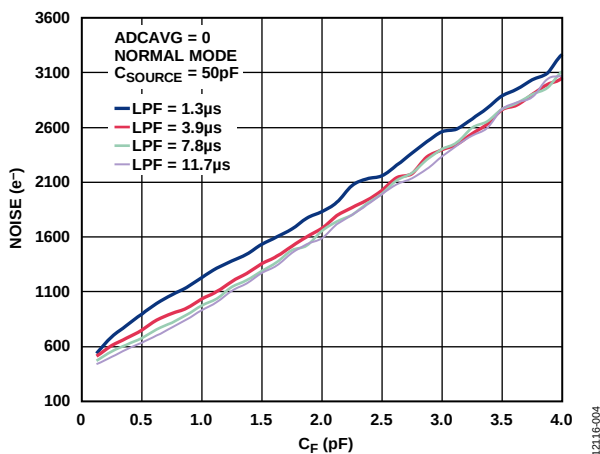
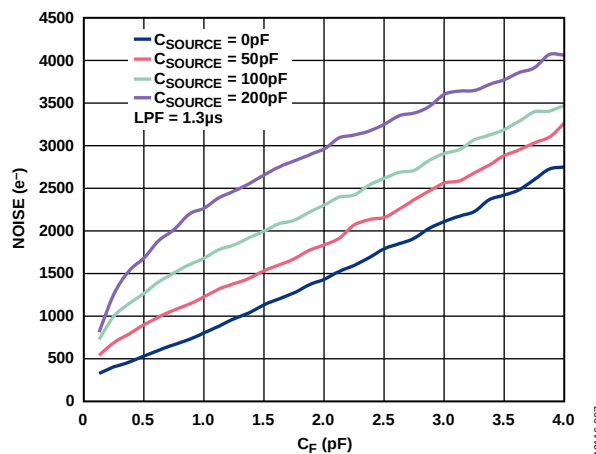
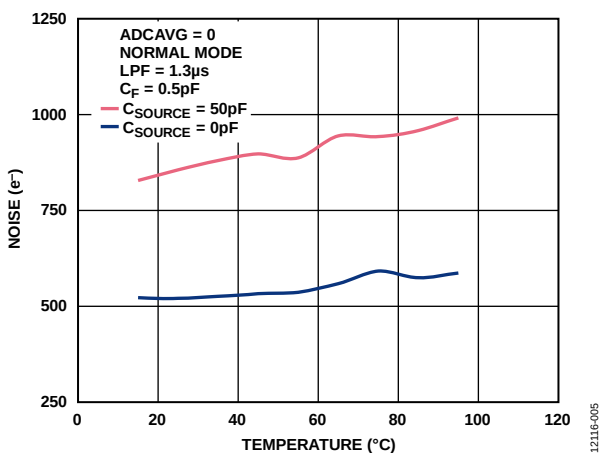
Figure 3. Noise vs. Integration Capacitor (C_F) for Various Power Modes, $ADCAVG = 0$ Figure 6. Noise vs. C_F for Various Power Modes, $ADCAVG = 1$ Figure 4. Noise vs. C_F for Various Low-Pass Filters (LPF)Figure 7. Noise vs. C_F for Various Source Capacitances (C_{SOURCE})

Figure 5. Noise vs. Temperature

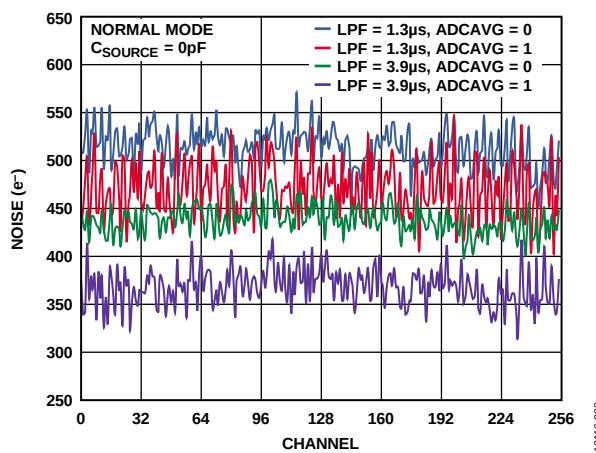


Figure 8. Noise vs. Channel

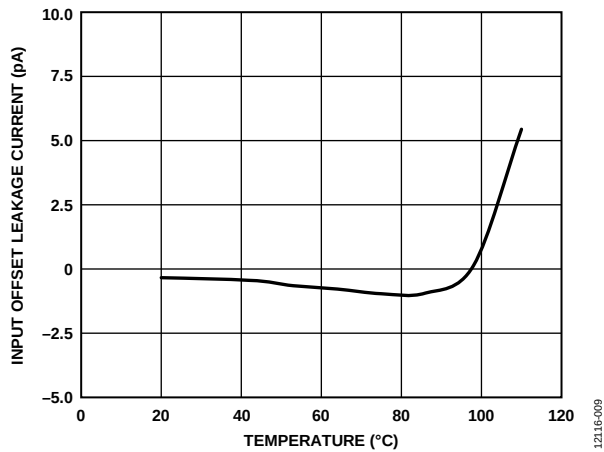


Figure 9. Input Offset Leakage Current vs. Temperature

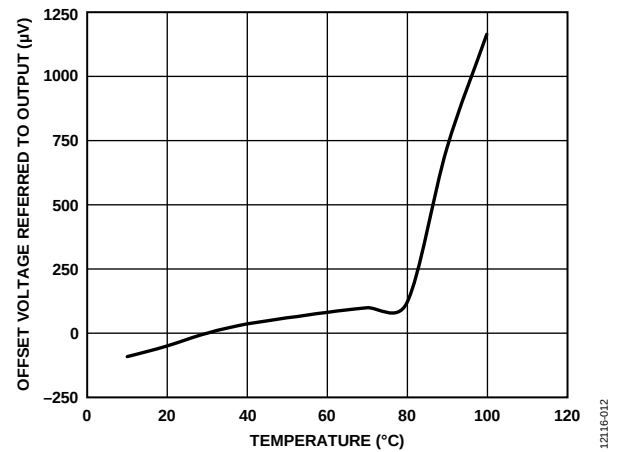


Figure 12. Total Offset Voltage Referred to Output vs. Temperature

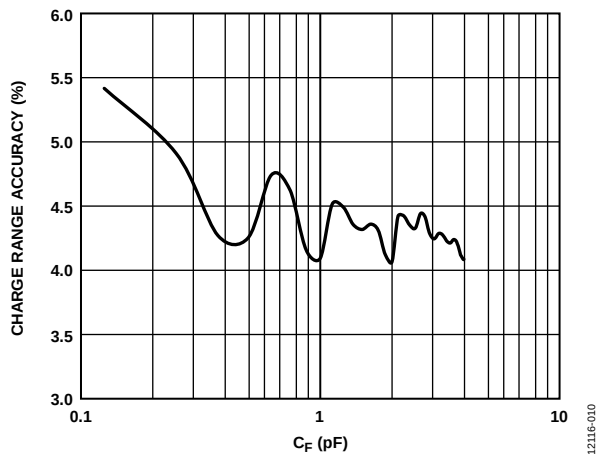
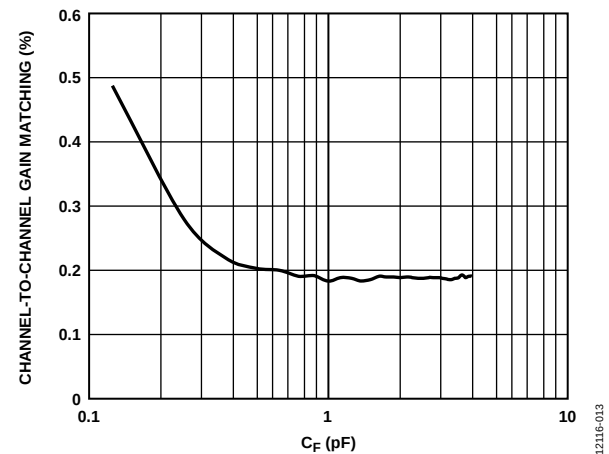
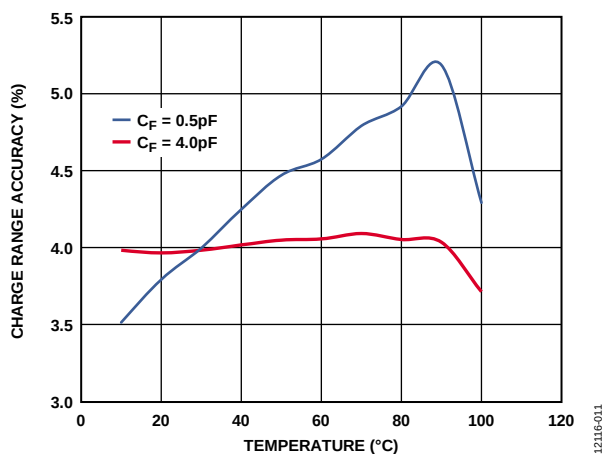
Figure 10. Charge Range Accuracy vs. C_F Figure 13. Channel-to-Channel Gain Matching vs. C_F 

Figure 11. Charge Range Accuracy vs. Temperature

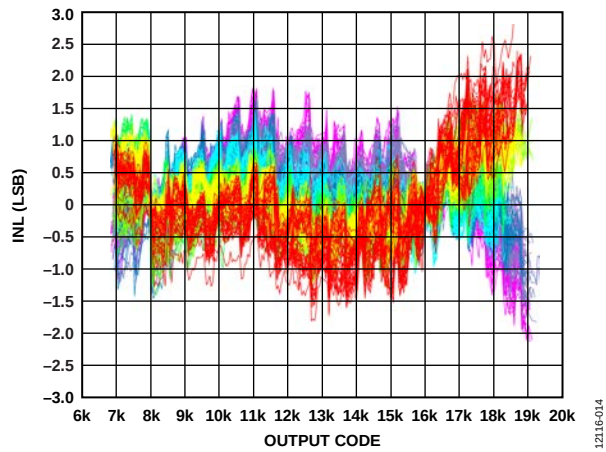


Figure 14. Integral Nonlinearity (INL)

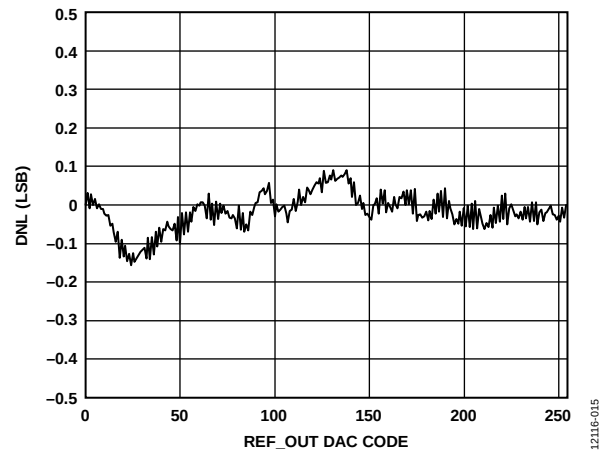


Figure 15. Differential Nonlinearity (DNL)

DEVICE OPERATION

The digital operation of the AD71143 is shown in Figure 19.

After power-up or a reset cycle, the configuration registers of the AD71143 can then be programmed via the synchronous SPI. The AD71143 has a power-on reset feature that puts the device in a known reset state, power-down mode. The SPI port consists of the CMOS signals: CS (Pin LD25), SCK (Pin LD26), SDI (Pin LD28), and SDO (Pin LD27). After the AD71143 is configured, it is ready for integration sampling, conversion operation, and data reading. These functions are controlled by the synchronous, self clocked serial interface that consists of the LVDS signals; the DCLKx inputs (Pin LD39 and Pin LD40); the DOUTAx and DOUTBx outputs (Pin LD33, Pin LD34 and Pin LD30, Pin LD31, respectively); and the CMOS signals: SYNC (Pin LD24) and ACLK (Pin LD23).

AFE OPERATION

The AFE of the AD71143 consists of a low noise amplifier in an integrator configuration, followed by an RC low-pass filter and a CDS block, as shown in Figure 18. The feedback capacitor of the integrator, which sets the full-scale range, is variable from 0.125 pF to 8 pF.

The feedback capacitor is controlled by the IFS bits (Bits[4:0] of Configuration Register 0). The time constant of the low-pass filter is also variable: from 1.3 μ s to 11.7 μ s. This value is controlled by the LPF bits (Bits[8:7] of Configuration Register 1).

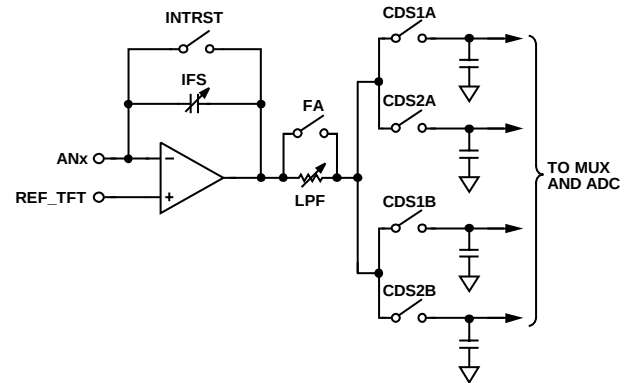


Figure 18. Simplified AFE Diagram

Several switches in the AFE open and close as defined by the user. The INTRST switch is used to reset the integrator to REF_TFT or the optional REF_INT. The FA switch is used to short the resistor of the low-pass filter for filter acceleration. The CDS switches connect the CDS capacitors to sample the output voltage of the integrator. The CDS1x capacitors sample the reset level of the integrator. The CDS2x capacitors sample the output voltage of the integrator. The difference between these two voltages is used to calculate the charge sampled by the integrator. The CDS stage has the added benefit of removing integrator offset, reset noise, and low frequency (1/f) noise.

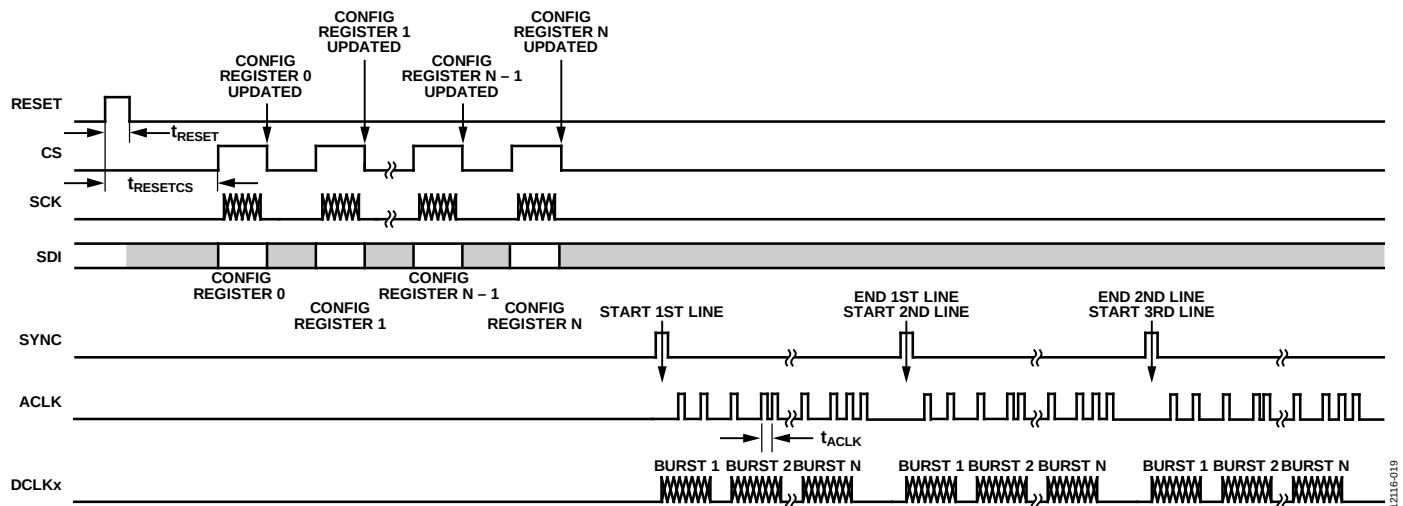


Figure 19. General Timing Diagram

Figure 20 shows the typical order of events in the sampling of a given line of a TFT panel. First, the integrator is reset to clear any charge from the previous line. Next, the reset level of the integrator is sampled onto the CDS1x capacitor. After this CDS1 activity, the charge is transferred from the TFT panel to the AFE by enabling the gates for a particular line on the panel. The output voltage of the integrator changes as a function of the charge transfer, and then this voltage is sampled onto the CDS2x capacitor. Finally, the ADC is ready to digitize the voltages stored on the CDS capacitors.

Figure 18 also shows the A and B pairs of the CDS capacitors. These two sets of capacitors are used in pipeline mode operation of the AD71143 (see the Conversion Modes section for an explanation of pipeline mode). After a reset, the first line sampling interval begins with the CDSxA capacitors. The CDSxB capacitors are used for the next line sampling interval. Bit 48 in the output data header indicates the pair of CDS capacitors used in the trailing data (see Table 11). In addition, writing the hexadecimal value of 0xFFFF to the SPI port ensures that the CDSxA capacitors are used in the next line being sampled.

Filter acceleration is useful if there are disturbances on the integrator or panel during the sampling process. By shorting the low-pass filter resistor with the FA switch, the CDS capacitors can settle more quickly to the output voltage of the integrator. The user can then open the FA switch so that the RC network can filter noise from the front end.

AFE Sequencer Timing

To control the order of the AFE sequencer timing events in the AD71143, the user must provide an ACLK signal and program a set of configuration registers, as shown in Table 5. The values in these configuration registers determine when switch activity occurs with respect to the various edges of an ACLK signal.

Table 5. AFE Configuration Registers

Switch	Configuration Register Address	Bits[7:4]	Bits[3:0]
INTRST	4	Close	Open
CDS1	5	Close	Open
CDS2	6	Close	Open
FA	7	CDS1 open	CDS2 open

ACLK is a pulsed, discontinuous clock from which all of the AFE timing is derived. The placement of clocks in the ACLK

pattern is defined by the user, which allows accurate placement of switch activity in the AFE.

The four user configuration registers shown in Table 5 program when switch activity (opening and closing) occurs with respect to the ACLK signal periods. The values programmed into the registers correspond to the rising edge of ACLK on which that activity occurs. There can be any number of ACLK periods in a line, but only 13 of them can correspond to switch activity in the AFE. Therefore, Bits[7:4] and Bits[3:0] of these configuration registers can have values from 0x0 to 0xD to correspond with one of these 13 ACLK periods.

It is recommended that the CDS2x capacitors be clear of charge from the previous line. When CDS2_RESETE is high (Bit 5 of Configuration Register 1), the CDS2 switch closes when both INTRST and CDS1 are closed. This resets the CDS2x capacitors (see Figure 21).

There may be instances (for example, during system debugging) when the user may want to have these switches always open or always closed. To set INTRST, CDS1, or CDS2 as always closed, set Bits[3:0] of their respective configuration register to 0xF. To keep these switches open, set Bits[3:0] to 0xE. If the 0xF or 0xE setting is used, Bits[7:4] are ignored.

The FA switch is unique in that it corresponds to a particular CDS stage. The closing of FA occurs automatically with the closing of CDS1 or CDS2. The opening of FA is controlled by Bits[7:4] after CDS1 closes, and by Bits[3:0] after CDS2 closes. Setting Bits[3:0] or Bits[7:4] to 0xF forces FA closed for the respective CDS stage. Likewise, setting these bits to 0xE keeps FA open for the respective CDS stage.

There are some timing recommendations for the AFE. If pipeline mode is selected, it is recommended that the first ACLK after a SYNC signal (ACLK0 in Figure 22) not be used to close the CDS1 switch. Instead, use the second ACLK after the SYNC signal (ACLK1 in Figure 22), which can occur one ACLK period after the first ACLK period (see Figure 22). It is recommended that the CDS time for CDS1 and CDS2 be equal to or greater than 5× the low-pass filter time constant, which is set by the LPF bits, Bits[8:7] of Configuration Register 1. See Table 6 for a list of recommended integrator reset times in normal mode. Low power mode integrator reset times are approximately five times longer.

Table 6. Integrator Reset Times (t_{INTRST}) for Various Panel Capacitances, No External Reset Voltage, Normal Mode

Panel Capacitance (pF)	Reset Time, $C_F \leq 2 \text{ pF}$ (μs)	Reset Time, $C_F > 2 \text{ pF}$ (μs)
0	6.0	12.0
70	6.0	12.0
80	6.0	15.0
90	6.2	16.2
100	6.6	17.7
120	10.0	22.2
150	16.0	30.0
200	43.2	64.0

PIPELINE = 0
CDS2_RESETEN = 0

SWITCH	BITS[7:4]	BITS[3:0]
INTRST	7	1
CDS1	0	3
CDS2	4	6
FA	2	5

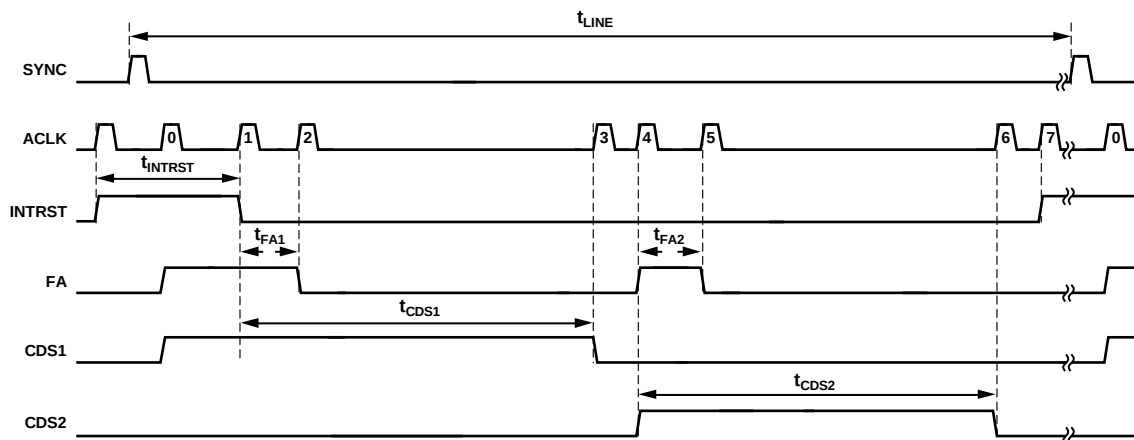


Figure 20. AFE Timing Example

PIPELINE = 0
CDS2_RESETEN = 1

SWITCH	BITS[7:4]	BITS[3:0]
INTRST	7	1
CDS1	0	3
CDS2	4	6
FA	2	5

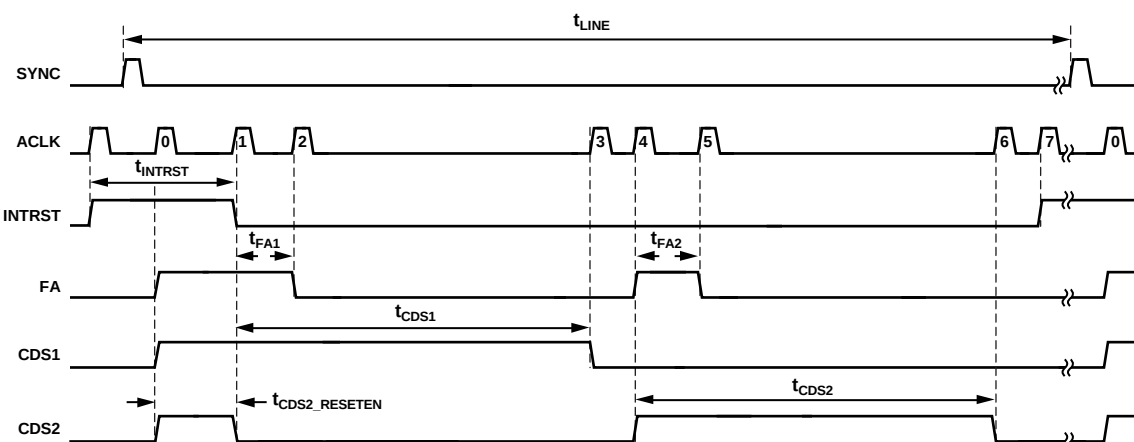


Figure 21. AFE Timing Example with CDS2 Reset Enabled

PIPELINE = 1
CDS2_RESETEN = 1

SWITCH	BITS[7:4]	BITS[3:0]
INTRST	8	2
CDS1	1	4
CDS2	5	7
FA	3	6

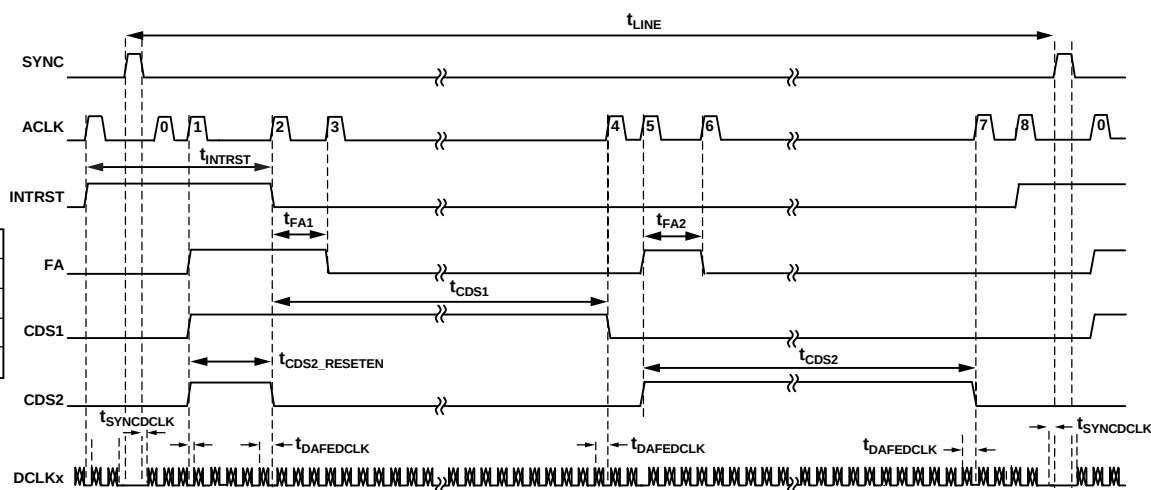


Figure 22. AFE Timing Example with Pipeline Mode Enabled

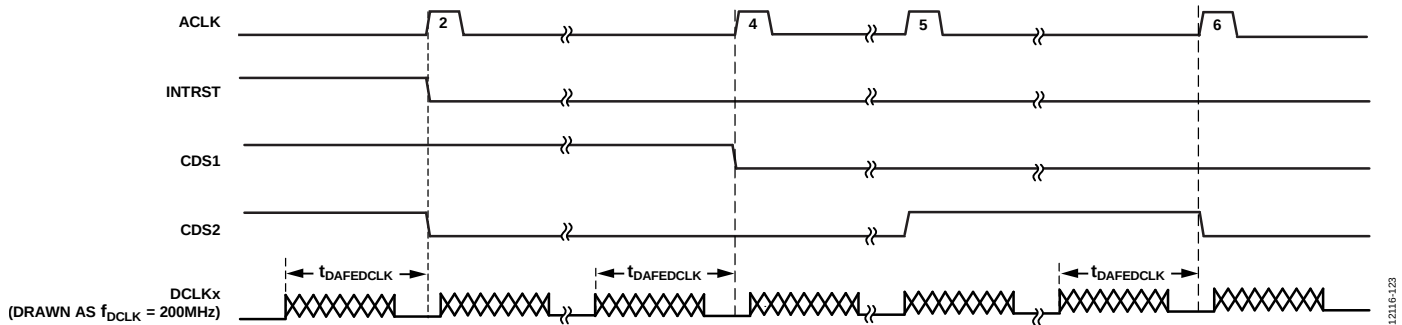


Figure 23. Detailed ACLK/DCLKx Timing with Pipeline Mode Enabled

Conversion Operation

The AD71143 conversion process and conversion data interface are controlled by three signals: ACLK, DCLKx (DCLKx in this data sheet refers to DCLKH and DCLKL), and SYNC.

The AFE sequencer clock, ACLK, is a pulsed discontinuous CMOS clock. ACLK is used to time a state machine, which controls the timing of the AFE. Each edge of ACLK is placed to correspond with a specific activity in the sampling process of the AFE. The number of ACLKs in a given line time is defined by the requirements of the user for AFE activity (see the AFE Operation section).

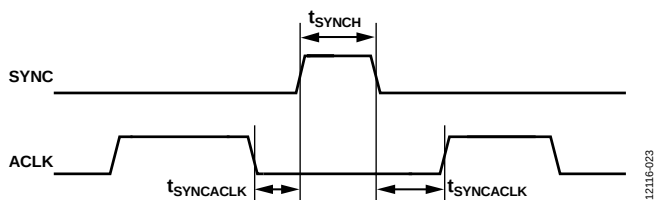


Figure 24. Detailed SYNC to ACLK Timing

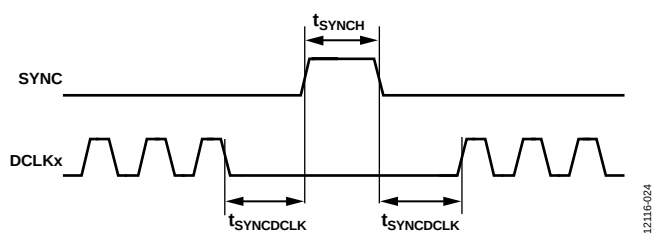


Figure 25. Detailed SYNC to DCLKx Timing

The data conversion clock, DCLKx, is a pulsed LVDS clock. The use of an LVDS clock minimizes the amount of wiring, reduces the synchronization challenge between multiple AD71143 devices and the digital host (FPGA), and helps reduce digital coupling to the analog inputs. DCLKx controls mux and ADC internal timing, as well as data output timing.

Both ACLK and DCLKx can be generated from the same master clock of an FPGA, which, in turn, is usually generated from a crystal oscillator. The typical period of ACLK, t_{ACLK} , is 20 ns; the typical period of DCLKx, t_{DCLK} , is 5 ns.

Unlike ACLK, DCLKx consists of a number of bursts within a conversion cycle. The total number of bursts depends on various data output modes, as described in the Data Interface section. Each burst contains 64 active clocks and a variable number of muted clocks (see Figure 26).

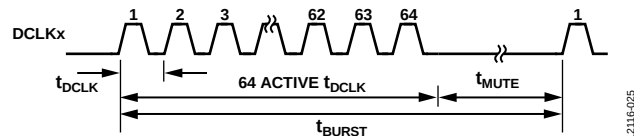


Figure 26. DCLKx Burst Diagram

The minimum burst length is 1765 ns when DCLKx is 200 MHz. This translates to a minimum mute time of 1445 ns.

Conversion Modes

The AD71143 provides modes that allow the user to determine when the conversion process takes place with respect to AFE sampling. These modes are controlled by the pipeline bit (Bit 0 of Configuration Register 2). When pipeline mode is selected by setting Bit 0 high, AFE sampling and the data conversion processes occur in parallel. When serial mode is selected by setting Bit 0 low, these two processes occur serially. The data conversion and data output processes always occur simultaneously; there is always a pipeline delay of 1 between them. Figure 27 shows the order of operation.

The use of pipeline or serial mode depends on the desired line time. In pipeline mode, the minimum line time is determined by whichever is longer: the AFE sampling time/ACLK activity (t_{AFE}) or the digital conversion time (t_{DCONV}). In serial mode, the minimum line time is the sum of t_{AFE} and t_{DCONV} . For details on AFE timing, see the AFE Sequencer Timing section. For digital conversion and data output timing, see the Data Interface section.

When in pipeline mode, avoid having DCLKx bursts overlap certain portions of the AFE sampling time. Specifically, there must be no DCLKx signals within 200 ns of the following switch activity: INTRST opening, INTRST closing, CDS1 closing, and CDS2 closing. The activity on these switches, particularly the closing of INTRST, CDS1, and CDS2, can disturb the ADC voltage conversion on the output of the integrator. See Figure 27 for a detailed timing example.

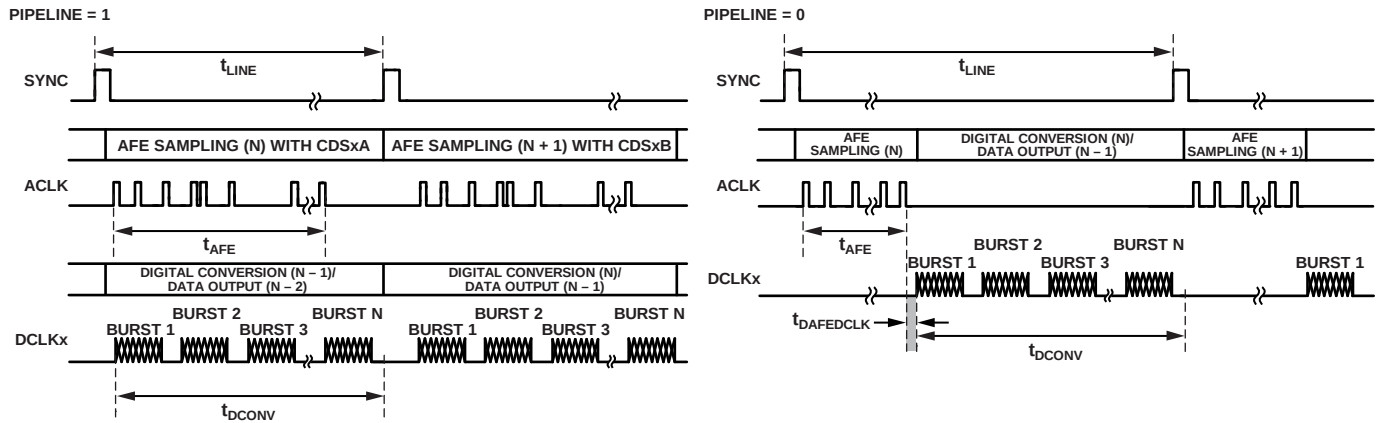


Figure 27. Comparison of Pipeline Mode and Serial Mode

Transfer Function

The channel charge results are output in 16-bit format in straight binary code, MSB first.

The LSB size of the charge in aC (attocoulomb, 10^{-18} C) is defined as

$$LSB_{CHARGE} = 0.125 \text{ (pF)} \times (IFS + 1) \times 61.035 \text{ (}\mu\text{V)}$$

where:

0.125 (pF) is the nominal feedback capacitance of the integrator.

IFS is the 5-bit Configuration Register 0 parameter that defines the full-scale range as set by the user.

61.035 (μ V) is the LSB size of the ADC.

For example, for IFS = 20, $LSB_{CHARGE} = 0.16$ fC.

The charge is calculated by the following:

$$Charge = (Data\ Out) \times LSB_{CHARGE}$$

where:

Data Out is the AD71143 channel output.

LSB_{CHARGE} is the LSB size previously calculated.

The full-scale charge range, FSR_Q , is

$$FSR_Q = 65,536 \times LSB_{CHARGE}$$

where 65,536 is the maximum number of LSBs.

The 16-bit temperature result is output on Bits[31:16] of the header (see Table 11). This temperature result can be used to monitor the change in temperature, but it is not recommended for use in measuring the absolute temperature. At 0°C, the nominal code is decimal 4380. The output typically changes at a rate of 17 LSB/°C. For example, if the temperature is 50°C, the temperature result is approximately 5230.

The voltage test input (VT) result is output on Bits[15:0] of the header (see Table 11). The weight of a VT LSB is 61.035 μ V. The VT result is relative to 0.5 V. For example, if VT = 1.0 V, the result is decimal 8192.

$$VTRESULT = (VT - 0.5)/61.035 \text{ }\mu\text{V}$$

Noise Considerations

The AD71143 provides four operational power modes, a sleep mode, and a power-down mode. The four operational power

modes control the current consumption and noise level in the integrator, allowing the user to make the trade-off between power and noise (see Table 7). The various power modes are selected via the PWR bits (Bits[8:6] of Configuration Register 0).

There is also a timing factor to consider. A 16-bit ADC requires 11 time constants for the signal to completely settle on its input. In the case of the AD71143, the integrated low-pass filter determines the time constant. Complete settling of the ADC results in better noise performance.

Table 7. Noise for Various Power Modes

Mode	Low-Pass Filter Time Constant (μ s)	Typical Noise, $C_{SOURCE} = 50$ pF (e^- rms) ¹	Typical Noise, $C_{SOURCE} = 100$ pF (e^- rms) ¹	Typical Total Power (mW)
Normal	1.3	885	1255	363
	7.8	660	1020	363
Low Power	1.3	925	1235	260
	7.8	805	1195	260

¹ The typical noise is measured using $C_F = 0.5$ pF, $T_J = 50^\circ\text{C}$. CDS times are $11 \times \tau_{LPF}$ for full 16-bit settling.

The AD71143 includes a feature that can be used to lower back-end noise, which is created by the CDS, the ADC buffer amplifier, and the ADC. For certain full-scale settings, this feature reduces total random noise. ADCAVG (Bit 1 of Configuration Register 1) and PIPELINE_AVGEN (Bit 0 of Configuration Register 10) enable this feature. In ADC averaging mode, the internal ADC averages the result of four consecutive ADC conversions, reducing noise.

This averaging increases ADC conversion, which requires more DCLKx bursts. A total of 130 bursts is required. There are 128 bursts for the integrators, one burst for VT and temperature, and one burst for the common-mode channel. The shortest possible line time in ADC averaging mode is 229.45 μ s.

Table 8. ADC Averaging Back-End Noise

ADCAVG	PIPELINE_AVGEN	Back-End Noise (μ V rms)
Low	Low	109
High	High	82

Power Optimization (Sleep Mode and Power-Down Modes)

The sleep mode and power-down modes of the AD71143 allow power savings without completely powering down the panel. Both the sleep mode and the power-down modes disable the back-end circuitry (mux, ADC, and digital processing). Sleep mode puts the integrator in a low power state, whereas the power-down modes completely disable the integrator.

The INTRST switch has a special relationship to sleep mode and the power-down modes, as shown in Table 9.

Table 9. Analog Inputs in Sleep and Power-Down Modes

Mode	PDTFTEN Bit ¹	INTRST Switch	Analog Inputs
Sleep	X	Closed	REF_TFT
	X	Open	Floating
Power-Down	Low	Closed	AVSSI
	High	Closed	REF_TFT
	X	Open	Floating

¹ X = don't care.

Sleep mode keeps the integrator functional so that the voltage that is present on the panel, REF_TFT, is maintained. The fast

switching time into and out of sleep mode, 0.1 ms, allows the user to conserve power in long frame rate applications where the AD71143 can toggle between sleep mode and one of the four higher power modes.

If INTRST is closed in sleep mode, the inputs are maintained to REF_TFT. If INTRST is closed in power-down mode, the inputs are shorted to AVSSI, the bottom rail of the integrator.

If INTRST is open in power-down mode, the inputs float. If the PDTFTEN bit (Bit 8 of Configuration Register 2) is high and INTRST is closed, the inputs are maintained to REF_TFT in power-down mode. This is the lowest power configuration for which the panel bias can be maintained.

The total power consumption can be further optimized with the REFDACDIS bit (Bit 7 of Configuration Register 2). When set high, REFDACDIS disables the on-board reference DAC, which is output on REF_OUT. Disabling the on-board DAC results in a 5 mW power savings. In a system configuration using multiple AD71143 devices, the REF_OUT voltage of one chip can be kept enabled and connected to the REF_OUT pins of all the other chips in the system. A similar scheme can be used when PDTFTEN is high in power-down mode.

Table 10. System Power Configurations for 10 AD71143 Devices

Mode	Bit Configuration ¹				Power (mW)	Total Power for 10 AD71143 Devices (W)	Test Conditions/Comments
	PWR[2:0]	LP_EN	REFDACDIS	PDTFTEN			
Power-Down	5	X	X	X	1.2	0.012	Device is powered down except for the SPI digital interface. Panel bias is maintained; REF_OUT is valid, with INTRST closed. REF_OUT of one AD71143 is connected to the REF_TFT and REF_INT pins of all AD71143 devices. There may be offsets among REF_TFT and REF_INT of the various AD71143 devices. Similar to power-down with panel bias optimized. Panel bias is maintained, and REF_OUT is valid.
Power-Down with Panel Bias	5	X	0	1	21	0.21	
Power-Down with Panel Bias Optimized	5	X	1	1	16	0.165	
Sleep Optimized	4	0	1	X	56	0.565	
Sleep	4	0	0	X	61	0.61	
Low Power	4	1	0	X	260	2.60	
Normal	0	X	X	X	363	3.63	

¹ X = don't care.

Power Supply Restrictions and Sequencing

There are some power supply restrictions to note on the AD71143. Some of these restrictions pertain to device operation, and some pertain to device power-up. All of the supplies in question are tied to the integrator of the AFE.

There are several voltages tied to the AFE that require special consideration during power-up. AVDDI is the top rail of the integrator input stage, and AVSSI serves as the bottom rail of the integrator input stage. AVDD5F serves as the top rail of the integrator output stage, and AVDD5Q serves as the quiet supply for the integrator. AGND5F and AGND5Q serve as the returns for these supplies. REF_TFT is the input voltage of the positive terminal of the integrator.

The critical relationship among all these voltages is shown in the following list. Note that these voltages differ from the operational conditions, shown in Figure 28 and Table 1.

- $0 \leq \text{AVDDI} - \text{AVSSI} \leq 2.7 \text{ V}$
- $|\text{AVDDI} - \text{REF_TFT}| \leq 2.7 \text{ V}$
- $|\text{AVSSI} - \text{REF_TFT}| \leq 2.7 \text{ V}$
- $\text{AVDD5F} - \text{AVSSI} \geq 0.0 \text{ V}$
- $\text{AVDD5Q} - \text{AVSSI} \geq 0.0 \text{ V}$

REF_TFT and REF_INT can be supplied by REF_OUT via the internal DAC, as shown in Figure 16. REF_OUT starts up at 2.5 V after all of the following voltages are applied: REF_ADC, AVDD2, AVDD5B, DVDD5, and DVDD2.

The user must ensure that all of these voltages power up before AVSSI. Figure 29 shows an example in which REF_OUT supplies both REF_INT and REF_TFT. Note that if AVSSI is at GND, there is no risk of supply sequencing issues. If REF_TFT is supplied externally, REF_TFT can be applied after REF_INT is set to its final value. In this case, AVDDI and AVSSI can be applied at the same time as REF_TFT.

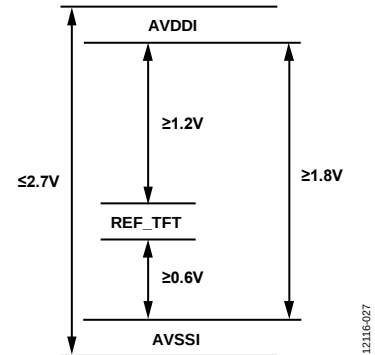


Figure 28. Operational AFE Supply Constraints

Figure 28 illustrates the relationship that must be maintained between AVDDI, AVSSI, and REF_TFT during operation. The difference between AVDDI and AVSSI must be between 2.7 V and 1.8 V. In operation, REF_TFT must be at least 1.2 V below AVDDI and at least 0.6 V above AVSSI.

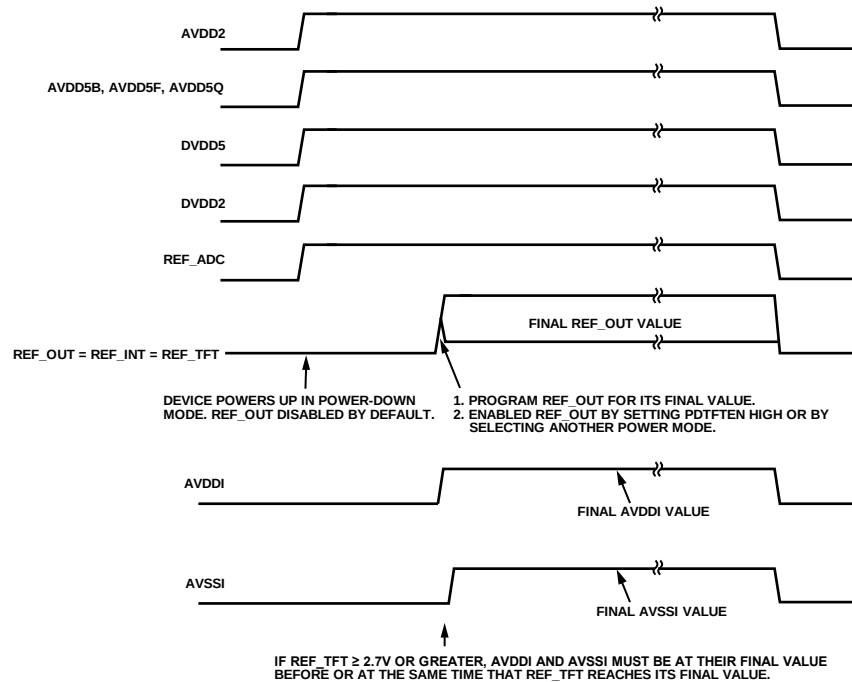


Figure 29. Power Sequencing Example, REF_TFT = REF_INT = REF_OUT

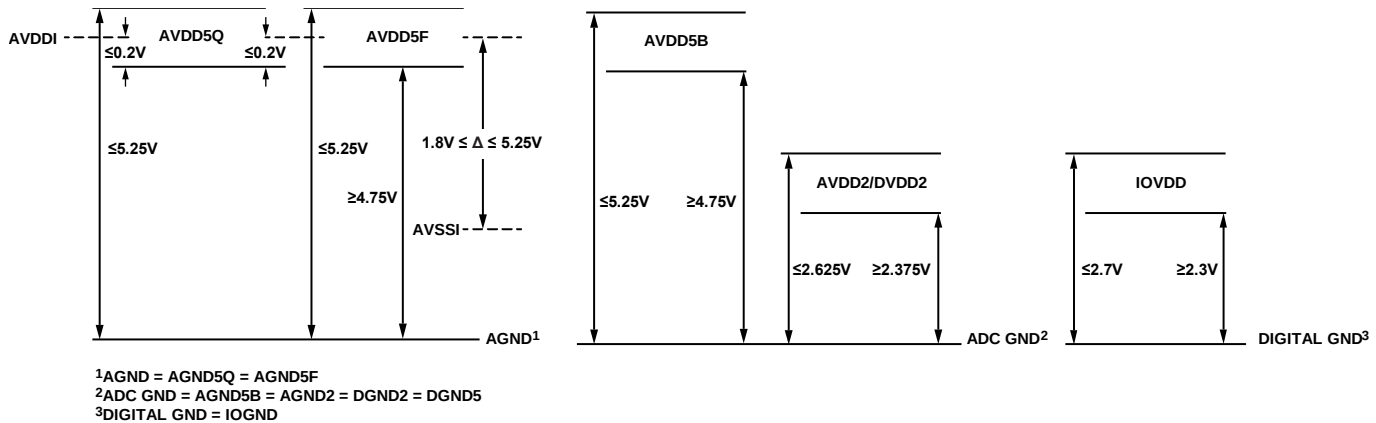


Figure 30. Supply Voltage Ranges

Reference Considerations

The AFE of the AD71143 can be configured in various ways to meet the needs of the user. The AFE provides two user-definable analog inputs. The first is REF_TFT, the integrator positive input voltage, which can be set to match the common-mode level of the panel of the user. The second is REF_INT, the integrator reset voltage, which can be set to partition the full-scale range of the integrator. REF_TFT can be any voltage from 0.6 V to 2.7 V and is typically defined by the panel requirements of the user. This is a very sensitive node that requires good decoupling. REF_INT can be any voltage from 0.5 V to 4.5 V.

Each integrator has a linear output range from 0.5 V to 4.5 V. Setting the REF_INT voltage within this span allows the user to partition the available range for positive and negative signals.

For example, in a holes integrating application, setting REF_INT to 4 V gives 87.5% of the available range for positive signals and 12.5% of the span for negative signals. Similarly, in an electron integrating application, setting REF_INT to 1 V partitions the span in the same way. Figure 31 illustrates this relationship.

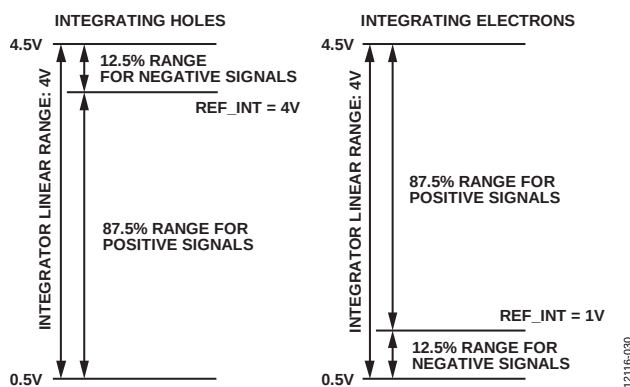


Figure 31. Integrator Linear Operating Range

REF_TFT = REF_INT = REF_OUT Configuration

There are three possible configurations for driving REF_TFT and REF_INT. The first and recommended configuration is to use the internal DAC to supply REF_TFT and REF_INT (see Figure 32). The internal reference DAC, which outputs on REF_OUT, is programmed by the REFDAC bits (Bits[7:0] of Configuration Register 3). The span of the DAC is from 0.5 V to 4.5 V, with an LSB size of 15.625 mV. Code 0 corresponds to 0.5 V, and Code 255 corresponds to 4.5 V. Note that REF_TFT must be in the range of 0.6 V to 2.7 V. Figure 32 illustrates the configuration that allows the AD71143 to easily track these critical voltages.

REF_TFT = External Source, REF_INT = REF_OUT

The second configuration allows the user to provide an external REF_TFT voltage of between 0.6 V and 2.7 V, and to connect REF_INT to REF_OUT (see Figure 33). This option is useful if the user does not want to connect REF_TFT to the same potential as REF_INT. For example, if the panel of the user integrates electrons and requires a 2.5 V REF_TFT, it may be desirable to have REF_INT at a lower voltage to maximize the available positive range. In this configuration, where REF_OUT is not driving REF_TFT, the user must set EXTRST (Bit 2 of Configuration Register 1) high.

REF_TFT = External Source, REF_INT = External Source

The third and least common configuration is to have the user provide both REF_TFT and REF_INT (see Figure 34). If the user supplies an external integrator reset voltage on REF_INT, the internal DAC must also be set to a value that closely matches the voltage on REF_INT. This is necessary for the AD71143 to properly calculate the CDS differentiation. If the external REF_INT voltage is 4 V, set the REFDAC bits (Bits[7:0] of Configuration Register 3) equal to Code 224 (0xE0). In this configuration, where REF_OUT is not driving REF_TFT, the user must set the EXTRST bit (Bit 2 of Configuration Register 1) high. Note that any discrepancy between an external REF_INT voltage and the REF_OUT voltage appears as an offset.

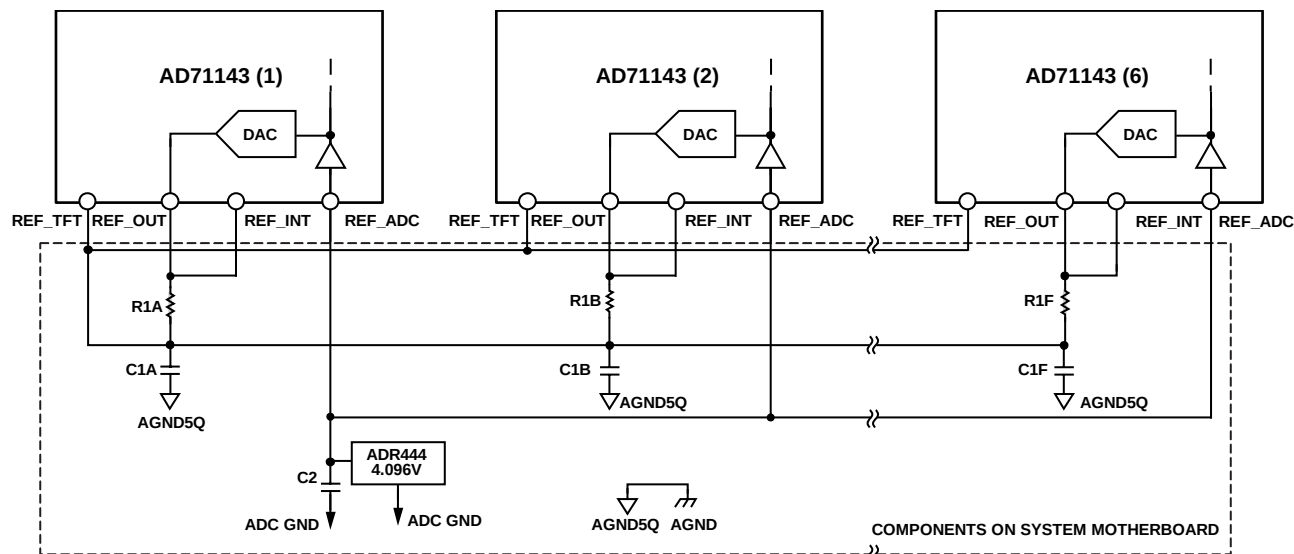


Figure 32. Reference Connection Diagram, REF_TFT = REF_INT = REF_OUT

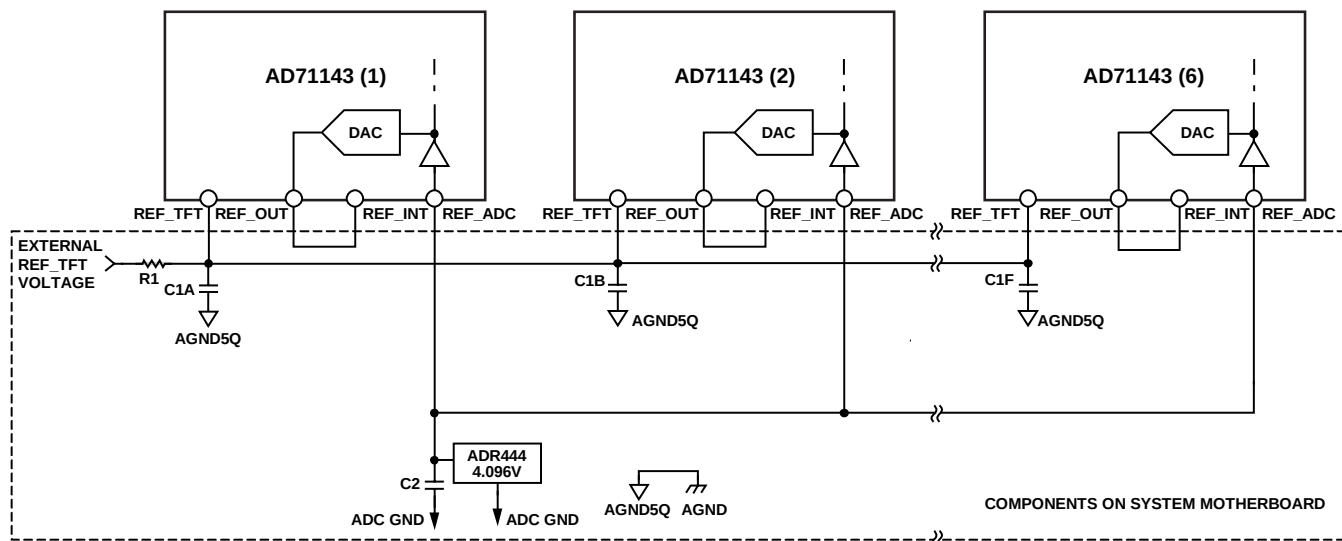


Figure 33. Reference Connection Diagram, REF_TFT = External Source, REF_INT = REF_OUT

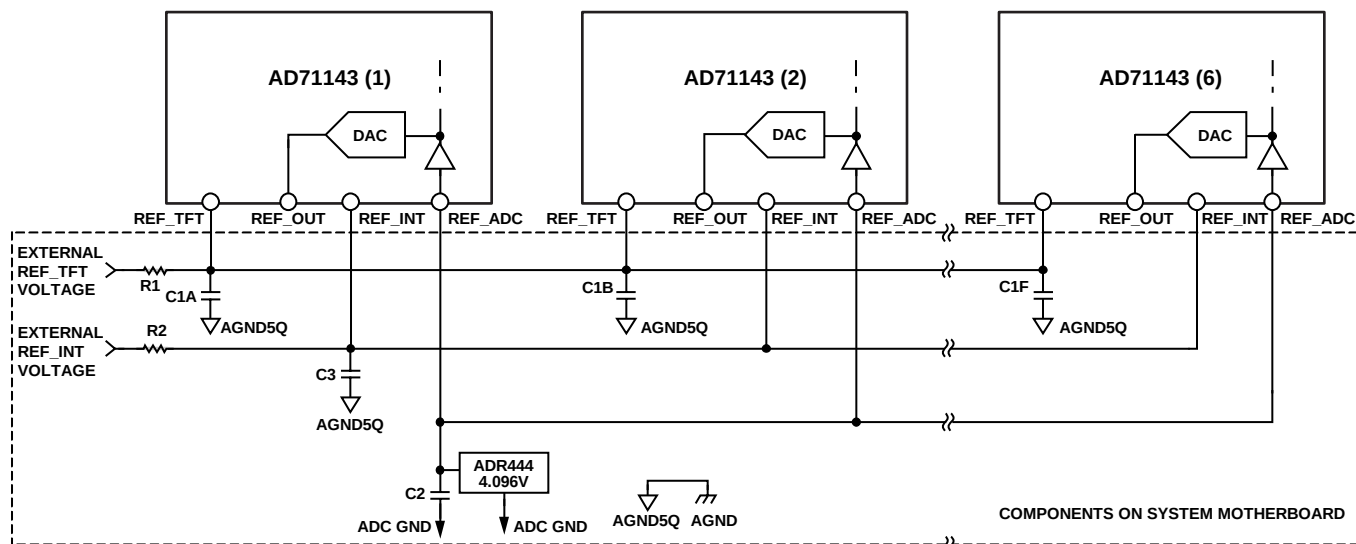


Figure 34. Reference Connection Diagram, REF_TFT = External Source, REF_INT = External Source

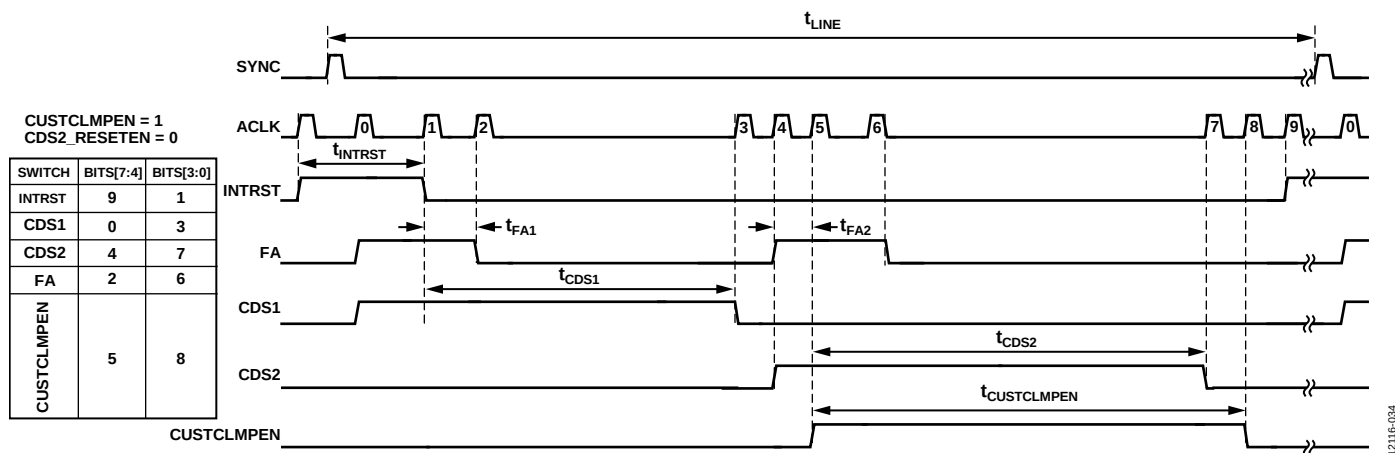


Figure 35. AFE Timing with Customized Integrator Clamping

Integrator Clamping

The AD71143 has an integrator clamp function that acts as an antiblooming mechanism. The integrator clamp detects when the output of the integrator is approaching saturation and, if necessary, removes the excess charge at the integrator input. The INTCLAMP bit (Bit 3 of Configuration Register 2) enables the integrator clamp functionality. The clamp is active from the falling edge of INTRST to the falling edge of CDS2. However, there is an option to customize the timing of this clamp using the AFE timing sequencer. This customizable clamp timing allows the user to create a window for use when the integrator clamp is active. This can be especially useful if the user wants to avoid a false clamping due to a large gate charge injection.

The customizable clamp timing is enabled by the CUSTCLMPEN bit (Bit 8 of Configuration Register 8). The edges of the time window are programmed in the same manner as other AFE timing. The CUSTCLMP_C bits (Bits[7:4] of Configuration Register 8) define the rising edge of the window, and the CUSTCLMP_O bits (Bits[3:0] of Configuration Register 8) define the falling edge of the window. The value of these bits corresponds to a particular ACLK. Note that the falling edge of CUSTCLMPEN must be after the falling edge of CDS2. Figure 35 shows an example using the customized integrator clamp timing.

The exact timing of this customizable integrator clamp depends on the AFE timing of the user, the fault conditions the user wants to detect, and the TFT panel characteristics.

Data Interface

The data interface of the AD71143 consists of two LVDS pairs: DOUTAx/DOUTBx and DCLKx/DCLKOx. The data lines (DOUTAx and DOUTBx) and the master clock (DCLKx) serve as the synchronous serial interface that is used to output the digital conversion results. DCLKOx is an echo version of

DCLKx that can ease data synchronization to the master clock. The data interface is highly customizable through various configuration register bits.

The DOUTMODE bit (Bit 2 of Configuration Register 2) allows the user to select between one of two data modes. In single data rate mode, when the DOUTMODE bit is low, the data is output on DOUTAx. In double data rate mode, when the DOUTMODE bit is high, the data is output on both DOUTAx (even channels) and DOUTBx (odd channels).

In general, following 34 DCLKx bursts, the user can begin to continuously clock DCLKx. This continuous clocking allows the user quicker access to data, independent of the line time. Furthermore, it defines the minimum line time that is achievable in pipeline mode: 70 μ s in single data rate mode and 60 μ s in double data rate mode. Note that, if a line time of 70 μ s is achieved in single data rate mode, there are insufficient DCLKxs to read out the configuration registers.

The READDOWN bit (Bit 3 of Configuration Register 1) controls the order in which data is output. When READDOWN is high, data starting with Channel 255 is output. Conversely, when READDOWN is low, data starting with Channel 0 is output. In either case, the results on DOUTx are always output MSB first.

Regardless of data mode or data output direction, there are some constants to the data pattern. The first burst of DCLKx always outputs a 64-bit header, as described in Table 11. The 256-channel results are output following 4096 clocks. If desired, additional clocks can output the contents of the configuration registers. After 4288 clocks, when all relevant data has been output (the header, channel data, and configuration register bits), 0s are output. Bringing SYNC high also forces DOUTx to 0. If a SYNC pulse occurs before the minimum number of bursts, the output data, conversion process, and AFE timing are corrupted.

Table 11. Header Descriptions

Header Bits	Description
[63:56]	The 8-bit fixed pattern; hexadecimal value: 0x0A.
[55:51]	These five bits are all low.
50	This bit is for factory use only.
49	Readout direction. If high, channel data starts with Channel 255. If low, channel data starts with Channel 0.
48	CDS identification. Useful in pipeline mode, this bit is high if data was sampled with CDSxA. If this bit is low, the data was sampled with CDSxB. This information is useful when trying to synchronize data from different frames.
[47:32]	These 16 bits are all low.
[31:16]	The 16-bit temperature result.
[15:0]	The 16-bit VT result.

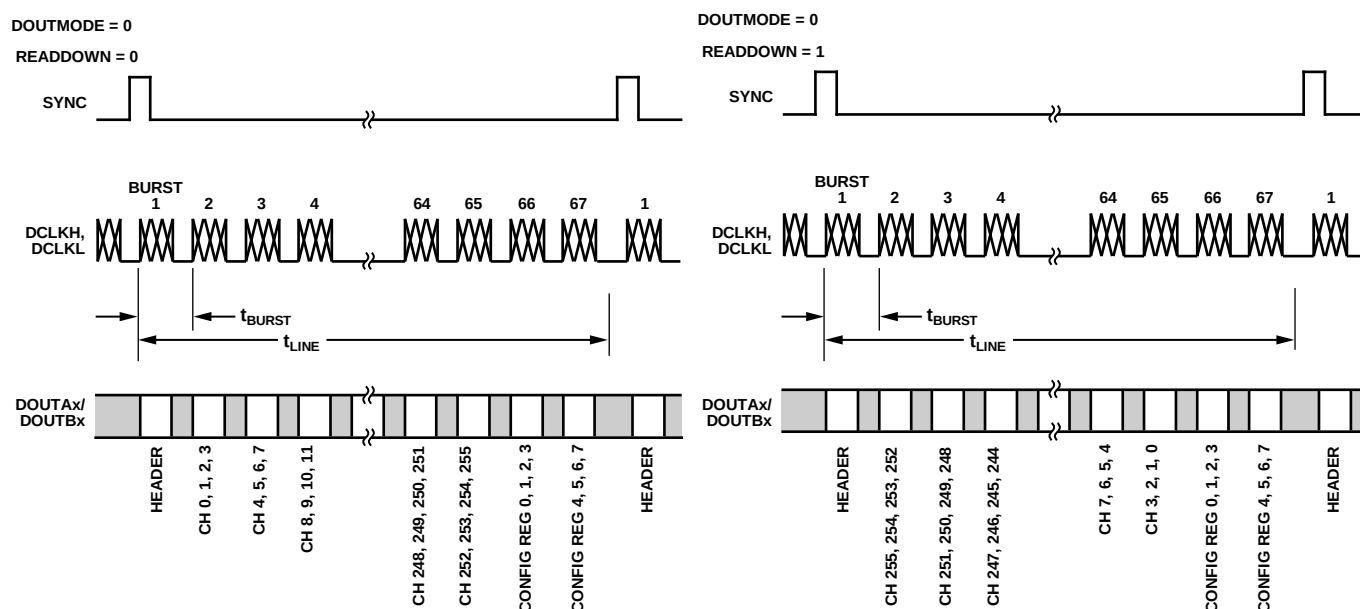


Figure 36. Data Output Format over a Single LVDS

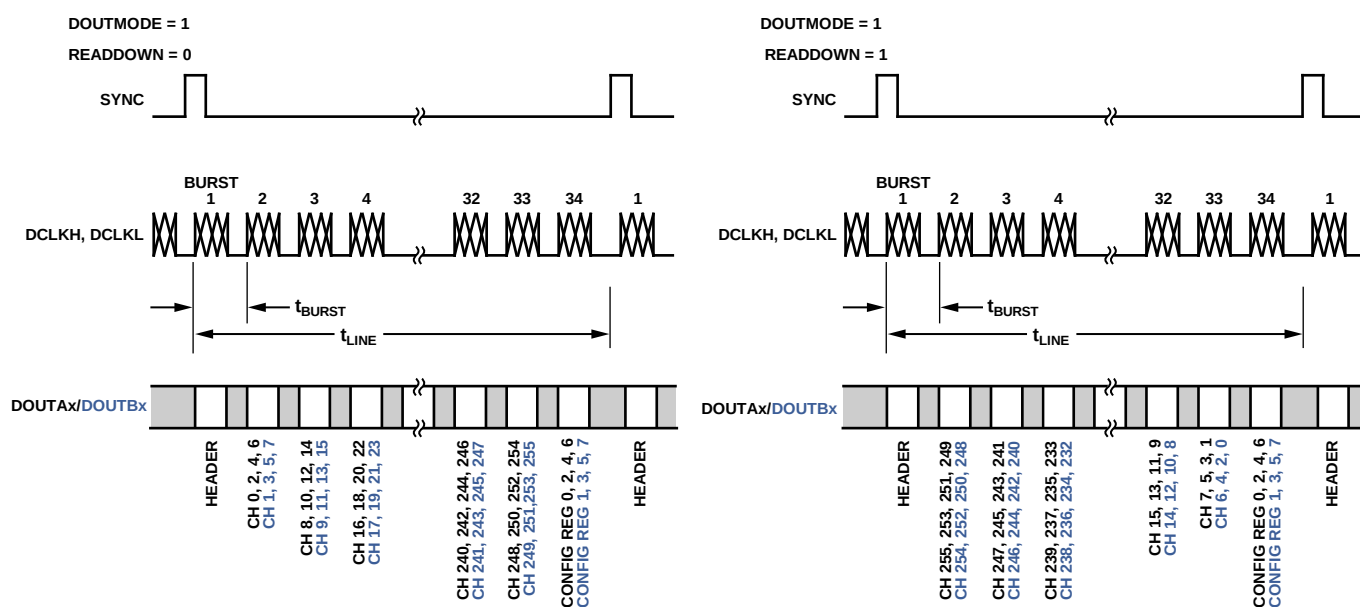
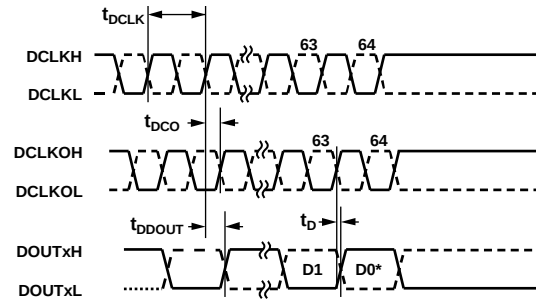


Figure 37. Data Output Format over a Dual LVDS

Echo Clock Interface Mode

The AD71143 offers an echo clock mode that is enabled by the ECHOCLK bit (Bit 1 of Configuration Register 2). This interface mode simplifies the synchronization of the data and the master LVDS clock. By setting ECHOCLK high, DCLKOx is enabled.

DCLKOx is a buffered copy of DCLKx and is synchronous to the data, DOUTAx or DOUTBx, which is updated on the rising edge of DCLKOx (t_{DCO}), as shown in Figure 38. By maintaining good propagation delay matching between DOUTxH, DOUTxL, and DCLKOx through the board and the digital host, DCLKOx can be used to latch the data, with good timing margin, into an FPGA.



*IF THERE IS MORE DATA TO OUTPUT, DOUTxH AND DOUTxL HOLD THE VALUE OF D0. IF THERE IS NO MORE DATA TO OUTPUT, DOUTxH AND DOUTxL GO TO 0.

Figure 38. Detailed LVDS Timing Interface

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Configuration Operation

The configuration of the AD71143 is defined by 16, 16-bit registers (Configuration Register 0 to Configuration Register 15). The functionality of each register bit is defined in Table 12. These registers are initialized to their reset value by a reset cycle (RESET is high).

The contents of the configuration registers are also output with the conversion data during each conversion cycle. This process can be used to monitor the integrity of the digital interface continuously.

The configuration registers can be updated via the 4-wire, SPI, synchronous serial interface with daisy-chain capability consisting of CMOS signals (CS, SCK, SDI, and SDO), as shown in Figure 39

and Figure 40. When multiple AD71143 devices are used, their SPI interfaces must be daisy-chained together.

A single write cycle can update any number of configuration registers in a chain of multiple AD71143 devices. The daisy-chain feature uses the SDI input and SDO output for cascading multiple AD71143 devices on a single data line that is similar to a shift register. This feature is useful for reducing component count and for wiring connections.

A daisy-chain connection diagram example using three AD71143 devices is shown in Figure 39, and the corresponding timing is shown in Figure 40.

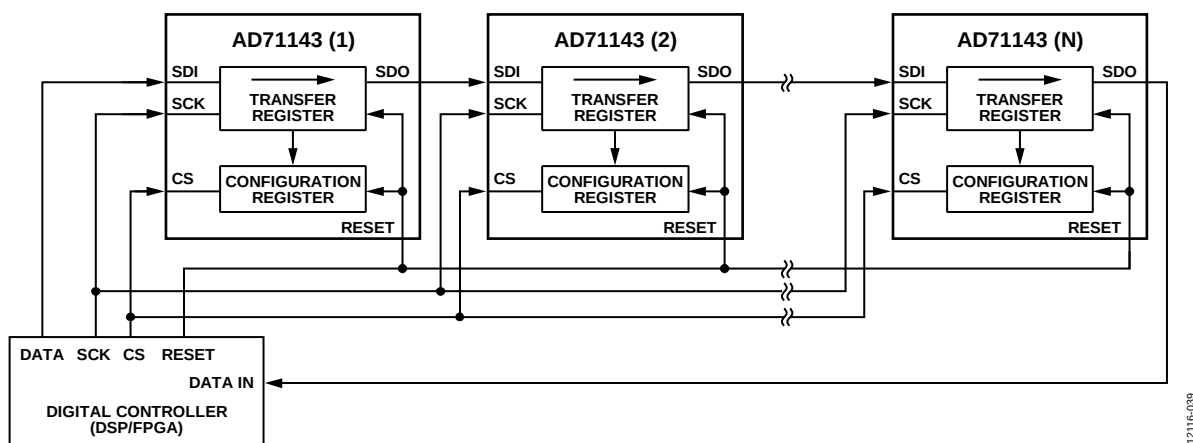


Figure 39. Daisy-Chain Configuration Connection Diagram

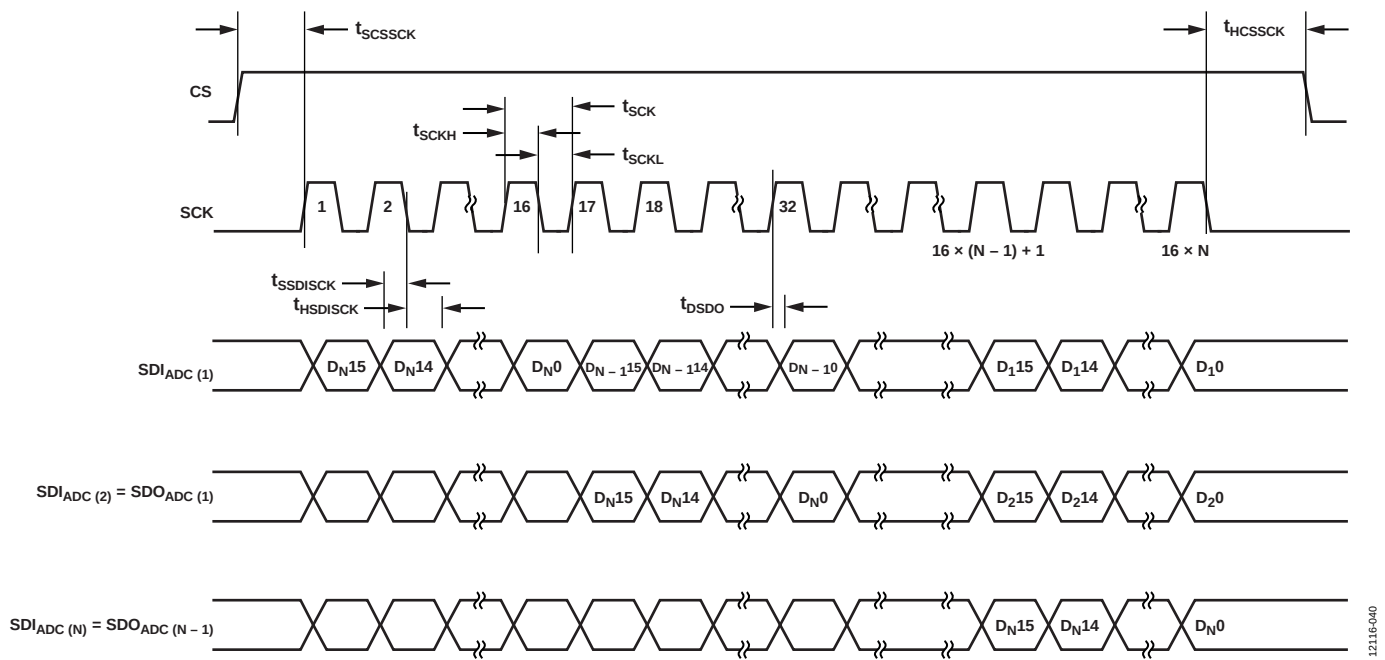


Figure 40. Daisy-Chain Configuration Timing Diagram

A write cycle into a configuration register is defined by a CS high pulse. The new value of the configuration register on the SDI input is shifted into an internal shift transfer register by the SCK clock, MSB first. SDI is sampled on the SCK falling edge. On the falling edge of CS, the configuration register is updated.

Also, during the write cycle, the data bits stored in the internal shift transfer register are clocked, MSB first, on the SDO output by the SCK rising edges.

During a write configuration operation, hold SYNC and DCLKx low.

During conversion, CS must be held low. SCK must also be held low, even though the SCK pulses are ignored.

To read configuration register contents from the SPI port, CS must be brought low for a minimum of two clocks after a read command has been issued and before the data can be read out of the SDO pin, as shown in Figure 41 (SCK 17 and SCK 18). The user can apply more than two clocks with CS low, if desired.

Digital Interface Integrity

The AD71143 has an internal linear feedback shift register (LFSR) that verifies the integrity of the digital interface. The values of the LFSR are read on DOUTAx and DOUTBx in the same manner as

conversion data (see Figure 36 and Figure 37). Each unique configuration of the AD71143 has a specific LFSR pattern, which allows the user to have integrity checks for different configurations. The LFSR is enabled by setting the LFSR_EN bit (Bit 6 of Configuration Register 11) high. The values of the LFSR can be reset either by applying a reset to the AD71143 or by writing 0xFFFE to the SPI port.

A typical implementation of the LFSR follows:

1. Configure the AD71143 via the SPI port as desired.
2. Enable the LFSR by setting the LFSR_EN bit high.
3. Read N number of lines.
4. Store the line results in a lookup table.
5. Compare the lookup table values to the results from a known working setup.

The results stored in the lookup table can be used to compare to other AD71143 devices. For a given configuration, each AD71143 yields the same values as those read in Step 3. If the user wants to check the digital integrity on a single AD71143, a sixth step that resets the LFSR can be added. Then Step 1 to Step 4 are repeated. Remember to disable the LFSR by writing 0xAC00 to the SPI port.

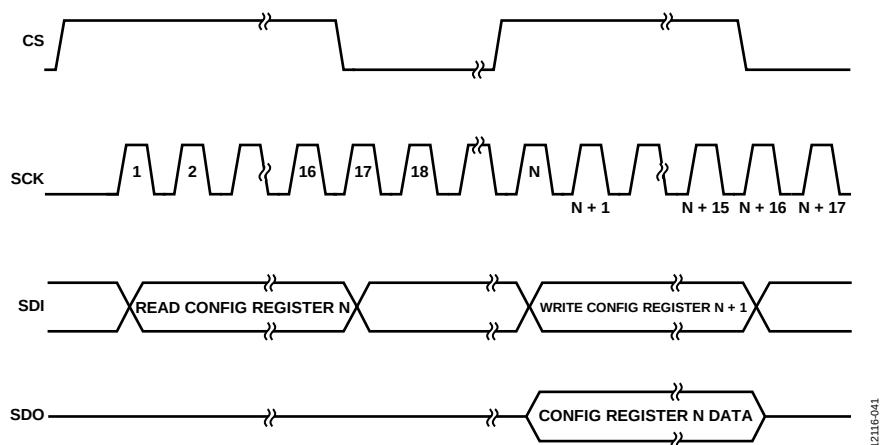


Figure 41. Reading a Configuration Register via SPI Port Timing

CONFIGURATION REGISTERS

Table 12. Configuration Register Contents

Reg. No.	Bit(s)	Bit Name	Description	Reset State	Readback Value	
0	15	Write	1: writes the 15 bits that follow in Configuration Register 0 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0	
	14	Read	1: the contents of Configuration Register 0 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0	
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 0 for Configuration Register 0.	0	0	
	9	Not used	This bit is for factory use only. Keep this bit low.	0	0	
	[8:6]	PWR	Power mode. These bits allow the user to optimize the power of the integrator for the desired noise level. Low power mode can be enabled by selecting sleep mode and then setting the LP_EN bit high (Bit 8 of Configuration Register 12).	5	PWR	
			PWR, Bits[8:6]			Power Mode
			0			Normal mode
			1			Unused
2			Unused			
3	Unused					
4	Sleep mode or low power mode with LP_EN = 1.					
5	Power-down mode					
5	Not used	This bit is for factory use only. Keep this bit low.	1	0		
[4:0]	IFS	Integrator full-scale selection. These bits allow the user to select the appropriate full-scale range. The full-scale range is between 0.5 pC and 16 pC.	0	IFS		
1	15	Write	1: writes the 15 bits that follow in Configuration Register 1 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0	
	14	Read	1: the contents of Configuration Register 1 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0	
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 1 for Configuration Register 1.	1	1	
	9	Not used	This bit is for factory use only. Keep this bit low.	0	0	
	[8:7]	LPF	Low-pass filter time constant. These bits allow the user to select the time constant of the low-pass filter at the output of the integrator. The time constant range is from 1.3 μs to 11.7 μs. With ADCAVG high, the time constant doubles.	0	LPF	
			LPF, Bits[8:7]			Time Constant (μs)
			0			1.3
			1			3.9
			2			7.8
	3	11.7				
6	Not used	This bit is for factory use only. Keep this bit low.	0	0		
5	CDS2_RESETEN	CDS2 reset enable. 1: forces the CDS2 switch to close when the INTRST and CDS1 switches are both closed. Allows the CDS2 capacitors to reset to REF_TFT when EXTRST is low, or to the external integrator reset voltage (REF_INT) when EXTRST is high. 0: the CDS2 capacitors are not reset; they maintain the voltage from their previous sample.	0	CDS2_RESETEN		
4	CMR_EN	Common-mode channel enable. 1: reduces common-mode errors from the AFE, such as power supply noise, temperature drift, or REF_TFT drift. The AD71143 contains eight additional integrators, identical to the other 256 integrators, except that their inputs are floating. The eight integrators are powered up, increasing AFE power by 1/32. During each line time, the result of these eight integrators is averaged together, and the measured error is then subtracted from the result of the other 256 integrators.	0	CMR_EN		

Reg. No.	Bit(s)	Bit Name	Description	Reset State	Readback Value
	3	READDOWN	Readout direction. Determines the order of the data readout. 1: the data readout order is from Channel 255 to Channel 0. 0: the data readout order is from Channel 0 to Channel 255.	0	READDOWN
	2	EXTRST	External integrator reset. 1: sets REF_INT to an externally supplied reset voltage. 0: resets the integrator to the REF_TFT voltage.	0	EXTRST
	1	ADCAVG	ADC averaging. 1: when the PIPELINE_AVGEN bit is high (Bit 0 of Configuration Register 10), the results of four consecutive ADC conversions are averaged. Averaging reduces the back-end noise from 109 μ V rms to 82 μ V rms.	0	ADCAVG
	0	Holes	Holes or electrons integration. This bit allows the AD71143 to always output an equivalent result in charge. 1: integrating holes. 0: integrating electrons.	0	Holes
2	15	Write	1: writes the 15 bits that follow in Configuration Register 2 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 2 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 2 for Configuration Register 2.	2	2
	9	Not used	These bits are for factory use only. Keep these bits low.	0	0
	8	PDTFTEN	Power-down TFT enable. 1: keeps the input stage of the integrator biased to REF_TFT when in power-down mode. With PDTFTEN high in power-down mode, the power-up time is 10 ms.	0	PDTFTEN
	7	REFDACDIS	Reference DAC disable. 1: allows the user to disable the on-board reference DAC, which saves 5 mW. This offers the user the option to customize power-down mode and sleep mode. The REF_OUT of one AD71143 can be used to set the REF_TFT and REF_INT of other AD71143 devices in a single panel.	0	REFDACDIS
	6	Not used	These bits are for factory use only. Keep these bits low.	0	0
	5	RNDOMIZE	Randomize. This bit enables random sampling of the 256 integrators via the internal multiplexer and helps minimize the effects of pattern noise. 1: random sampling. 0: the sampling process is performed in a sequential manner, starting with Channel 0.	1	RNDOMIZE
	4	Not used	This bit is for factory use only. Keep this bit low.	0	0
	3	INTCLAMP	Integrator clamp. 1: enables internal clamping circuitry on the input of the integrators. This prevents the integrator from saturating to its rails by removing any excess charge present on the integrator. This feature acts as an antiblooming mechanism.	0	INTCLAMP
	2	DOUTMODE	Data output mode. This bit enables double data rate mode. 1: data is output on both the DOUTAx and DOUTBx pins. 0: serial mode. The conversion data is output serially on the DOUTAx pins.	0	DOUTMODE
	1	ECHOCLK	This bit enables DCLKOx, the optional LVDS echo clock output.	0	ECHOCLK
	0	Pipeline	This bit allows the user to determine when the conversion process takes place with respect to the AFE sampling. 1: AFE sampling and the data conversion process occur in parallel (pipeline mode). 0: AFE sampling and the data conversion process occur serially. There is always a pipeline delay of 1 in the data output with respect to the conversion process.	0	Pipeline

Reg. No.	Bit(s)	Bit Name	Description	Reset State	Readback Value
3	15	Write	1: writes the 15 bits that follow in Configuration Register 3 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 3 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 3 for Configuration Register 3.	3	3
	9	Not used	This bit is for factory use only. Keep this bit low.	0	0
	8	AZEN	Auto-zero enable. 1: enables the auto-zeroing function, which offers an improvement in 1/f noise. VT and temperature are not digitized by the ADC. There is no data associated with VT and temperature; 0s are output.	0	AZEN
	[7:0]	REFDAC	Reference DAC. These bits determine the value for the internal reference from 0.5 V to 4.5 V. LSB size = 15.625 mV.	128	REFDAC
4	15	Write	1: writes the 15 bits that follow in Configuration Register 4 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 4 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 4 for Configuration Register 4.	4	4
	[9:8]	Not used	These bits are for factory use only. Keep these bits low.	0	
	[7:4]	INTRST_C	Integrator reset switch close control. These bits control when the timing integrator reset switch closes. The value of these bits corresponds to the rising edge of ACLK in a frame. Acceptable values for INTRST_C are 0x0 to 0xC.	7	INTRST_C
	[3:0]	INTRST_O	Integrator reset switch open control. These bits control when the timing integrator reset switch opens. The value of these bits corresponds to the rising edge of ACLK in a frame. A value of 0xF keeps this switch closed. A value of 0xE keeps this switch open. When 0xF or 0xE is used, INTRST_C is ignored. Acceptable values for INTRST_O are 0x0 to 0xC, 0xE, and 0xF.	1	INTRST_O
5	15	Write	1: writes the 15 bits that follow in Configuration Register 5 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 5 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 5 for Configuration Register 5.	5	5
	[9:8]	Not used	These bits are for factory use only. Keep these bits low.	0	
	[7:4]	CDS1_C	CDS1 switch close control. These bits control when the CDS1 switch closes. The value of these bits corresponds to the rising edge of ACLK in a frame. Acceptable values for CDS1_C are 0x0 to 0xC.	0	CDS1_C
	[3:0]	CDS1_O	CDS1 switch open control. These bits control when the CDS1 switch opens. The value of these bits corresponds to the rising edge of ACLK in a frame. A value of 0xF keeps this switch closed. A value of 0xE keeps this switch open. When 0xF or 0xE is used, CDS1_C is ignored. Acceptable values for CDS1_O are 0x0 to 0xC, 0xE, and 0xF.	3	CDS1_O
6	15	Write	1: writes the 15 bits that follow in Configuration Register 6 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 6 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 6 for Configuration Register 6.	6	6
	[9:8]	Not used	These bits are for factory use only. Keep these bits low.	0	
	[7:4]	CDS2_C	CDS2 switch close control. These bits control when the CDS2 switch closes. The value of these bits corresponds to the rising edge of ACLK in a frame. Acceptable values for CDS2_C are 0x0 to 0xC.	4	CDS2_C

Reg. No.	Bit(s)	Bit Name	Description	Reset State	Readback Value
	[3:0]	CDS2_O	CDS2 switch open control. These bits control when the CDS2 switch opens. The value of these bits corresponds to the rising edge of ACLK in a frame. A value of 0xF keeps this switch closed. A value of 0xE keeps this switch open. When 0xF or 0xE is used, CDS2_C is ignored. Acceptable values for CDS2_O are 0x0 to 0xC, 0xE, and 0xF.	6	CDS2_O
7	15	Write	1: writes the 15 bits that follow in Configuration Register 7 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 7 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 7 for Configuration Register 7.	7	7
	[9:8]	Not used	These bits are for factory use only. Keep these bits low.	0	
	[7:4]	FA_CDS1	Filter acceleration switch for CDS1. These bits control the filter acceleration switch for the low-pass filter of CDS1. The value of these bits corresponds to the rising edge of ACLK in a frame. A value of 0xF keeps this switch closed, and a value of 0xE forces this switch open. Acceptable values for FA_CDS1 are 0x0 to 0xC, 0xE, and 0xF.	2	FA_CDS1
	[3:0]	FA_CDS2	Filter acceleration switch for CDS2. These bits control the filter acceleration switch for the low-pass filter of CDS2. The value of these bits corresponds to the rising edge of ACLK in a frame. A value of 0xF keeps this switch closed, and a value of 0xE forces this switch open. Acceptable values for FA_CDS2 are 0x0 to 0xC, 0xE, and 0xF.	5	FA_CDS2
8	15	Write	1: writes the 15 bits that follow in Configuration Register 8 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 8 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 8 for Configuration Register 8.	8	8
	9	Not used	These bits are for factory use only. Keep these bits low.	0	0
	8	CUSTCLMPEN	Custom clamp enable. 1: enables the custom clamp timing feature.	0	CUSTCLMPEN
	[7:4]	CUSTCLMP_C	Custom clamp closed. These bits control when the custom clamp comparator is enabled, activating the clamp circuitry. The value of these bits corresponds to the rising edge of ACLK.	0	CUSTCLMP_C
	[3:0]	CUSTCLMP_O	Custom clamp open. These bits control when the custom clamp comparator is disabled, deactivating the clamp circuitry. The value of these bits corresponds to the rising edge of ACLK.	0	CUSTCLMP_O
9	15	Write	1: writes the 15 bits that follow in Configuration Register 9 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 9 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 9 for Configuration Register 9.	9	9
	[9:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	7	7
10	15	Write	1: writes the 15 bits that follow in Configuration Register 10 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 10 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 10 for Configuration Register 10.	10	10
	[9:1]	Not used	These bits are for factory use only. Keep these bits low.	0	0
	0	PIPELINE_AVGEN	Pipeline average enable. 1: enables ADC averaging during pipeline mode.	0	PIPELINE_AVGEN

Reg. No.	Bit(s)	Bit Name	Description	Reset State	Readback Value
11	15	Write	1: writes the 15 bits that follow in Configuration Register 11 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 11 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 11 for Configuration Register 11.	11	11
	[9:7]	Not used	These bits are for factory use only. Keep these bits at their reset value.	0	0
	6	LFSR_EN	Linear feedback shift register enable. 1: enables the LFSR feature. See the Digital Interface Integrity section for details.	0	LFSR_EN
	[5:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	24	24
12	15	Write	1: writes the 15 bits that follow in Configuration Register 12 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 12 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 12 for Configuration Register 12.	12	12
	9	Not used	This bit is for factory use only. Keep this bit low.	0	0
	8	LP_EN	Low power mode enable. 1: enables low power mode when sleep mode is enabled by setting PWR bits to 4.	0	LP_EN
	[7:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	2	2
13	15	Write	1: writes the 15 bits that follow in Configuration Register 13 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 13 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 13 for Configuration Register 13.	13	13
	[9:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	35	35
14	15	Write	1: writes the 15 bits that follow in Configuration Register 14 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 14 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 14 for Configuration Register 14.	14	14
	[9:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	43	43
15	15	Write	1: writes the 15 bits that follow in Configuration Register 15 on the next CS falling edge. 0: ignores the 15 bits that follow. For the write operation to work, the read bit must be low.	0	0
	14	Read	1: the contents of Configuration Register 15 are output on SDO after the next rising edge of CS (see Figure 41). For the read operation to work, the write bit must be low.	0	0
	[13:10]	ADDR	Configuration register address bits. ADDR must be set to 15 for Configuration Register 15.	15	15
	[9:0]	Not used	These bits are for factory use only. Keep these bits at their reset value.	8	8

OUTLINE DIMENSIONS

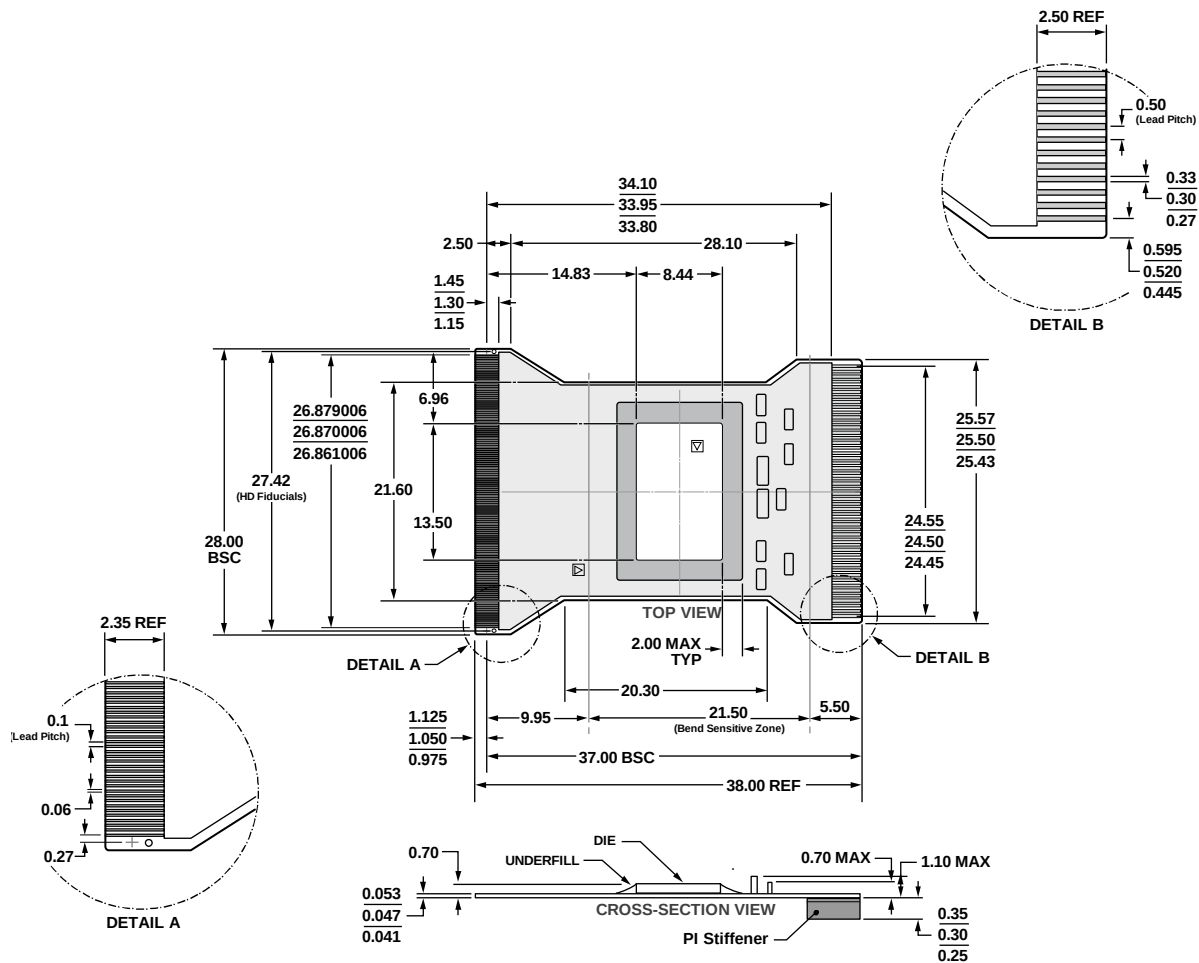


Figure 42. 320-Lead System-on-Film [SOF]
(CF-320-5)
Dimensions shown in millimeters

ORDERING GUIDE

Model ^{1,2}	Temperature Range	Package Description ²	Package Option
AD71143JCFZ	0°C to 70°C	320-Lead System-on-Film [SOF]	CF-320-5

¹ Z = RoHS Compliant Part.

² Contact Analog Devices for information about other SOF models.